Natural Disasters and Early Child Development: Evidence from an Earthquake*

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Abstract

This paper evaluates the persistent impact of the 2010 Chilean earthquake on children's cognitive and non-cognitive development. Using a difference-in-differences strategy that exploits quasi-random spatial and birth-cohort variation, I track children exposed between the prenatal period and age four over a seven-year period, comparing them with children born after the earthquake. I find that early-life exposure leads to significant and enduring declines in both cognitive and non-cognitive test scores seven years later, with suggestive evidence of short-term cognitive disruption. The adverse cognitive effects are particularly pronounced among boys in both the short- and medium-term. To explain these deficits, I investigate a multidimensional disruption of the early caregiving environment. First, the disaster triggered a sharp increase in maternal smoking participation and stress. Second, I provide suggestive evidence of a structural asset shock: I observe a short-term reduction in household income per capita and maternal labor force participation. While these indicators recovered within two years, exposure to higher seismic intensity is associated with significant private expenditures on home repairs. Taken together, these results are consistent with a stress-budget trap: while households preserve the quantity of formal human capital investments, the evidence points to a combination of compounding stressors and a reallocation of resources away from the unobserved quality of the domestic environment to finance physical recovery. These findings suggest that early-life developmental penalties can be triggered by acute behavioral responses to trauma and the temporary dismantling of the domestic buffer, persisting even after household income and institutional access return to baseline.

Keywords: Natural Disasters, Earthquake, Early Childhood, Cognitive Development, Non-cognitive Development, Smoking

JEL Codes: I25, O1, Q54, I120

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1 Introduction

Natural disasters pose a significant threat to human development. Over the past few decades, their frequency and severity have increased (E. J. Kim 2016) resulting in negative consequences across a wide range of economic outcomes. The immediate impact undermines healthcare and educational facilities, damage infrastructure, and disrupt water and sanitation systems, while also having a direct impact on the mental health of those affected, and in the worst cases, losses of life. The medium- and long-term effects can hinder human capital accumulation, health (Weldeegzie 2017), labor market outcomes (Gignoux and Menéndez 2016), and overall economic growth (Heger and Neumayer 2019).

A growing body of literature indicates that the negative effects of natural disasters on human capital accumulation are especially pronounced for individuals exposed during the prenatal period or early childhood (Almond et al. 2018, Torche 2011, Neelsen and Stratmann 2011, Menclova and Stillman 2020). This high vulnerability arises because skill formation is an accumulative process; a person's current skill level is heavily influenced by past levels (Cunha and Heckman 2007, Currie and Almond 2011). Consequently, shocks occurring in critical early stages (such as prenatal development or infancy) can adversely affect both cognitive and non-cognitive development. As a result, these early disruptions not only impair a child's current skills but also have lasting implications for long-term outcomes in adulthood, including educational attainment and wages (Akresh et al. 2012, Currie and Rossin-Slater 2013, Aizer et al. 2016, Shah and Steinberg 2017, Paudel and Ryu 2018). Moreover, negative shocks tend to disproportionately affect children from low-income backgrounds, as their families often lack the financial and social resources needed to buffer against such adversities. As a result, these children face greater challenges in recovering from early setbacks, which can hinder their developmental trajectories and long-term outcomes.

Identifying the causal effects of negative shocks during early childhood is challenging due to potential confounding factors related to both observed and unobserved family characteristics. To address this issue, several studies have used both human-induced shocks and natural disasters as sources of exogenous variation in early life conditions. These include exposure to violence (from internal and external conflicts, Camacho 2008, Akresh et al. 2012, Quintana-Domeque and Ródenas-Serrano 2017, Duque 2017, Kreif et al. 2022), excessive rainfall (Rosales-Rueda 2018, Sajid and Bevis 2021, Aguilar and Vicarelli 2022), hurricanes (Currie and Rossin-Slater 2013, Oliveira et al. 2021), and earthquakes (Torche 2011, Caruso and Miller 2015, Paudel and Ryu 2018). Notably, earthquakes stand out from other natural disasters because they fulfill several unique criteria regarding unpredictability; their timing and location are inherently difficult to anticipate, this sharply limits concerns that exposure is correlated with anticipatory behaviors, selective mobility, or other endogenous responses that complicate identification in settings such as floods, storms, or heat shocks.

In this paper, I evaluate the impact of the 2010 Chilean earthquake on key developmental indicators in children. This earthquake, which registered a magnitude of 8.8 on the moment magnitude scale, is ranked as the 6th strongest earthquake ever recorded. To estimate its effects,

I combine the classification from Astroza et al. 2010 to define municipalities impacted by the earthquake with data from the Early Childhood Longitudinal Survey (ELPI), to assess the short- and medium-term adverse consequences on both cognitive and non-cognitive development.

To this end, as an identification strategy, I leverage on the plausibly exogenous seismic intensity across the territory and use two sources of identification commonly used in the literature (Akresh et al. 2012, Weldeegzie 2017, Rosales-Rueda 2018, Aguilar and Vicarelli 2022). First, I exploit cohort variation by comparing children conceived before the earthquake, who were in utero and up to aged four at the time of the event, with those conceived afterward. The latter group must serve as a valid counterfactual to control for unobservable differences across municipalities. This requires that unobserved heterogeneity at the local level remain constant across birth cohorts and is adequately captured by year-of-birth fixed effects. Second, I exploit geographic variation by comparing children in municipalities that experienced more severe earthquake impacts to those in less impacted areas. Any cohort-related differences in unobserved characteristics must be comparable between impacted and non-impacted municipalities, which are accounted for through municipality fixed effects.

I find that seven years after the tremor, children in utero and those up to four years old experienced a decrease of 0.17 standard deviations (SDs) in cognitive performance measured by the Peabody test. This reduction is economically significant; this translates into an average loss of 0.8 years of schooling for children in the most affected municipalities at the moment that these children are measured (D. Evans and Yuan 2019). In terms of non-cognitive development, in the medium-term, I find a reduction of 0.13 SDs. In the short-term, I also find 0.14 SDs decrease on cognitive development, though this effect is marginally significant at conventional levels and does not survive multiple hypothesis testing correction, indicating that the primary evidentiary contribution rests on the medium-term estimates. The heterogeneity analysis unmasks that boys are relatively more adversely affected by the earthquake in terms of cognitive development but not on non-cognitive development. Additionally, I explore other dimensions that traditionally determine a household's capacity to buffer external shocks such as education and age of the parents and the presence of the father in the household finding no difference in development. The main findings are robust to other intensity measures such as non-linear peak ground acceleration and distance to the epicenter, as well as the inclusion of municipality-specific linear trends and regional fixed effects. I further validate the identification strategy through placebo and falsification tests. Crucially, event-study estimates support the parallel trends assumption, showing no systematic differences in outcomes for cohorts born after the earthquake. Taken together, these checks suggest that the observed developmental setbacks are driven by the seismic shock itself rather than pre-existing trends or omitted variables.

I explore three sets of mechanisms to explain the persistence of these developmental deficits. First, I examine birth outcomes and I find a statistically significant but economically modest reduction in height at birth (-0.38 cms), with no significant impact on birth weight or gestational age for children who were in utero. These results suggest that the mid-term damage is not primarily driven by traditional neonatal health shocks.

Second, I investigate household resource allocation and human capital inputs. While the earthquake triggered a sharp 10% reduction in short-term per capita income and a 5 percentage point drop in maternal labor supply, these effects dissipated within two years. Moreover, I find no significant impact on standard health investments, such as breastfeeding, vaccination rates, or medical check-ups and daycare or school enrollment. This suggests that while the institutional quantity of care was preserved, the domestic buffer was diminished. Leveraging the exogenous physical variation in seismic intensity, I provide an account of a structural asset shock: households in high-intensity municipalities were 10 percentage points more likely to finance major home repairs using their own funds, at a cost equivalent to a full month of income per capita. Although cross-sectional, these estimates are supported by balance tests showing that PGA intensity is uncorrelated with 2009 baseline income, education, or house quality, suggesting the shock was effectively random.

Third, I analyze maternal risky behaviors for mother of children from in utero and up to six months-old as a proxy for earthquake-induced trauma. Mothers in the most affected areas were 8 percentage points more likely to behave in risky ways, specifically, 4.6 percentage points more likely to smoke, a 41% relative increase that is nearly eight times larger than the behavioral response typically elicited by a major cigarette tax hike (W. N. Evans and Ringel 1999). Notably, this response is driven by the extensive margin and occurs despite tightened household budgets. Complementary evidence confirms that this behavioral shift was accompanied by a significant rise in reported stress which was specifically amplified among mothers who were actively parenting young children during the disaster. As noted by DeCicca et al. 2022, the marginal harm of smoking to fetal health is highest at low levels of consumption, where even minor increases can trigger hypoxia and impair development (Lakdawala and Simon 2017).

I interpret these risky behaviors not as the sole mechanical cause of the deficits, but as visible biomarkers for a broader trauma bundle that disrupted the primary caregiving environment. These findings imply a potential trade-off: while households successfully preserved the quantity of formal investments, such as schooling and vaccinations, the diversion of liquid assets toward physical recovery and the surge in maternal distress likely forced a sacrifice in the unobserved quality of home-care. This multifaceted disruption provides a consistent explanation for both the biological scarring among infants and the structural resource scars that impacted the broader toddler cohort.

Finally, I conduct a series of robustness checks to address potential confounding factors. Using Vital Record data and the ELPI survey, I find no evidence of selective fertility or unintended pregnancy shifts that would bias the composition of the treated cohorts. Similarly, migration was minimal and insufficient to explain the observed developmental gaps. To address a 6% differential attrition rate in impacted municipalities, I employ longitudinal survey weights and demonstrate that the main estimates remain robust to the bounding techniques of Lee 2009 and Kling et al. 2007. These results suggest that the identified human capital deficits are a direct consequence of the disaster rather than a byproduct of non-random sample selection or demographic shifts.

This paper contributes to the literature on early-life shocks and child development by providing evidence on the impact of a major earthquake during an understudied developmental window. While

several studies examine the effects of natural disasters on children's cognitive or non-cognitive skills, virtually none focus on earthquakes. These events stand out because their timing is inherently unpredictable, which provides an unusually robust source of exogenous variation in exposure compared to more seasonal shocks.

Existing earthquake-based studies tend to concentrate on immediate impacts on birth outcomes (Torche 2011, Menclova and Stillman 2020, Álvarez-Aranda et al. 2020), physical health (Bustelo et al. 2012), or terminal indicators like years of schooling that remain unchanged once they have materialized (Paudel and Ryu 2018, Shidiqi et al. 2023). The only comparable study focusing on cognitive development is Andrabi et al. 2021, who documented negative effects on test scores four years after the 2005 earthquake in Pakistan. To the best of my knowledge, no study to date has documented the effects of a major earthquake on non-cognitive development. I fill this gap by examining both cognitive and non-cognitive skills in the medium term, seven years after the 2010 Chilean earthquake.

This specific time-frame is crucial, yet largely absent in the literature, because early skill gaps begin to crystallize into persistent developmental trajectories during this window, while public interventions can still prevent permanent long-term losses (Almond et al. 2018). Measuring outcomes in this period offers a timely view of the mechanisms through which early shocks translate into later disadvantage; it provides policymakers with an actionable horizon that short-term physiological measures or final educational attainment cannot capture. Furthermore, by focusing on school-aged children, researchers and institutions can leverage the school system as a centralized infrastructure for assessment and treatment. This approach significantly reduces the logistical costs associated with the household-level data collection required for younger cohorts, making the medium-term window not only theoretically vital but also a more cost-efficient entry point for large-scale policy monitoring and intervention.

Second, this study contributes to the growing literature on the effects of natural disasters on risky behaviors as coping mechanisms for high-stress situations (DiMaggio et al. 2009, Do and Iyer 2012, Ayyagari and Sindelar 2010). Specifically, I extend this literature by examining how these behaviors, triggered during the sensitive stages of pregnancy and early infancy, influence child development. Crucially, by identifying that this shift occurs almost exclusively on the extensive margin, I highlight a particularly damaging pathway; as noted by DeCicca et al. 2022 and Lakdawala and Simon 2017, the marginal harm to fetal health is highest at precisely the low-consumption levels triggered by this initiation and relapse. This moves the debate beyond simple income effects to show how environmental trauma programs lower developmental trajectories through objective, measurable behavioral shifts.

Finally, this study addresses a gap in the human capital literature. While the link between maternal risky behaviors and physical health is well-documented (Nilsson 2017, Kreif et al. 2022), evidence regarding cognitive and non-cognitive development remains sparse. This paper builds on work by Srivastava and Trinh 2021 on environmental shocks and test scores by demonstrating how such trauma manifests through observable behavioral markers. By identifying maternal smoking

as a proxy for a wider internal environment shock, the study suggests that the cumulative impact of behavioral shifts and domestic instability, rather than an isolated biological channel, drives the observed developmental penalties.

This paper proceeds in Section 2 where I describe the 2010 Maule Earthquake in Chile. In Section 3, I present the data, the descriptive statistics, the empirical strategy, and the validity tests for the identification strategy. Section 4 shows the results in the short- and medium-term, heterogeneity analysis, and robustness checks. Section 5 suggests possible mechanisms, and finally, Section 6 summarizes the main findings and concludes.

2 The 2010 Maule Earthquake

Chile is situated along the boundary between the Nazca and South American plates, one of the most seismically active subduction zones in the world. On February 27, 2010, at 03:34 a.m., an M_w 8.8 mega-thrust earthquake struck central Chile. It remains the second strongest in Chilean history and was, at the time, the world's sixth largest recorded seismic event. The resulting rupture generated a catastrophic tsunami with run-up heights reaching 29 meters, impacting over 500 km of coastline (Larrañaga and Herrera 2010)[1].

The scale of physical destruction was immense, directly affecting two million residents and resulting in 547 confirmed fatalities. Approximately 550,000 housing units were severely affected, with 370,000 homes deemed uninhabitable. Infrastructure damage was widespread: one-third of the educational facilities in the affected regions were damaged, and over 70% of the national hospital network experienced some level of impact (Reconstrucción 2015). Critical utilities (electricity, water, and communications) were severed for periods ranging from days to weeks, disproportionately harming low-income and rural communities (OPS 2010).

Crucially, the earthquake was not a singular event but a prolonged exposure. The affected regions experienced thousands of aftershocks in the subsequent month (March 2010), including several exceeding M_w 6.0 (López Tagle and Santana Nazarit 2011). These episodes sustained high levels of maternal stress and uncertainty. While the government launched the Reconstruction Plan 2010, providing over 240,000 housing subsidies, the recovery of private dwellings was notably slow; by 2011, only 24% of damaged homes had been delivered (Comerio 2013). This persistent housing instability eventually led to the 2014 "Reconstruction Closure Plan" which explicitly addressed the psychosocial consequences of the disaster mitigating trauma and social dislocation among households whose lives remained disrupted years after the event.

In contrast, the institutional infrastructure for human capital recovered rapidly. Assessments by CEPAL 2010 indicate that 92.7% of health facilities suffered only minor or no damage, with most becoming fully operational by June 2010. Furthermore, structural damage was highly concentrated, with 44% of affected facilities located in the Maule region.[2] Schools and public daycare centers

[1] A more detailed analysis of the background of the earthquake is available in the appendix A

[2] Regressions excluding the Maule Region in Table C.29 in the appendix yield results consistent with the main specification.

resumed operations by April and May 2010, respectively. This rapid recovery of services meant that children in the 2011 cohort were born into a functional healthcare and educational environment. This distinguishes them from the 2006–2010 cohorts, whose critical developmental windows coincided with the acute phase of the shock: severe maternal stress, home destruction, and the immediate breakdown of local support systems.

Economically, the disaster cost roughly 18% of Chile's GDP, with affected municipalities experiencing a 10% reduction in economic activity that persisted nearly a decade later (Aguirre et al. 2023). These figures underscore why large scale earthquakes warrant attention: while infrequent relative to other hazards, their destructive potential is catastrophic[3]. Evidence suggests that the full economic cost of such shocks unfolds over decades through fetal and early childhood exposure, making the 2010 Maule earthquake a critical setting for identifying the medium- to longer-term consequences of environmental shocks on human capital.

Figure 1 shows the impacted regions that happen to be the central regions of the country and the Peak Ground Acceleration intensities (PGA) at the municipality level[4]. According to Astroza et al. 2010 the regions affected were between 5th and 9th regions including the Metropolitan region[5]. The remaining regions although they felt the shock to some extent, they did not suffer any major damage.

To complement the contextual previously presented, which characterize the 2010 earthquake as one of the most catastrophic events in Chile's modern history, I provide evidence based directly on the micro-data, I document the intensity of the shock through its impact on psychological and physical infrastructure.

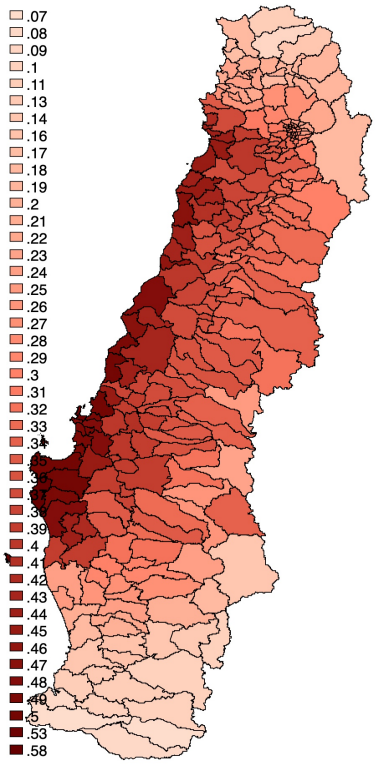
Using the CASEN Earthquake panel data, in Table 10, both the earthquake binary indicator based on Astroza et al. 2010 and the continuous PGA measure are strongly and significantly correlated with increased maternal stress (measured by the Davidson Trauma Scale) and physical destruction. Furthermore, Table C.13 documents the human toll, showing that fatalities were highest in the regions where the PGA was most intense (e.g., Maule and Biobío). These results suggest that the earthquake did not merely cause temporary alarm in a prepared population; it fundamentally disrupted the domestic environment through house destruction and significant psychological strain. This evidence supports the argument that the observed developmental setbacks are driven by a combination of material loss and a stress shock that persisted well beyond the immediate event.

[3]Globally, earthquakes of magnitude 7.0-7.9 occur approximately 10-20 times per year, while events of magnitude 8.0 or higher occur, on average, once annually (Earthquake Hazards Program 2023)

[4]In the next section, Peak Ground Acceleration will be defined. In the appendix, the Figure D.1 extends the map for the whole country

[5]By 2010, Chile was structured with 15 regions, with the formation of the 16th region (Ñuble) occurring later in 2017. Therefore, I stick to the 2010 definition. Additionally, there are 179 municipalities representing the 345 municipalities in Chile. The number of earthquake tagged municipalities is 143 and the share of the earthquake municipalities is 79%. This aligns with the official stats that state that 80% of the population was impacted by the earthquake.

Figure 1: Intensities of the 2010 Chilean Earthquake at Municipality Level



Note: The figures displays the Peak Ground Acceleration intensities in the impacted regions.

3 Data and Empirical Strategy

3.1 Data

The first set of data comes from the Early Child Development Survey (ELPI for its acronym in Spanish, 2012 and 2017 waves) which is a nationally representative survey of Chilean children born between 2006 and up to 2016. In each wave, ELPI adds new children to have a representative sample of all children in the country[6]. This survey contains detailed socio-demographic, employment, and income information of the members of the family. Notably, it contains rich data on parenting and cognitive, non-cognitive, and anthropometric indicators of the child that are aimed to measure development at different stages of their life. Furthermore, key health and birth outcomes (weight, size, gestational age, etc.) were transcribed from the objective 'Personal Child Health Record' documents.[7].

In this paper, I use the Peabody Picture Vocabulary test (Peabody) to assess cognitive development and the Child Behavior Check List (CBCL) to measure non-cognitive development[8]. Additionally, I use the Battelle Developmental Inventory - Screening test (Battelle) which integrates cognitive and non-cognitive dimensions. These tests are normalized using Chilean, Hispanic or international norms. In Table C.1 in the appendix, I summarize the birth cohorts and the age at which they took each test.

The Spanish-language version of the Battelle Developmental Inventory Screening test assesses the achievement of developmental milestones in children aged 7 to 84 months. Specifically, it evaluates communication, social-emotional, adaptive, motor, and cognitive skills. It is important to note that, as a screening tool, this test is not intended for detailed analysis by sub-sections.

In contrast, the Peabody Picture Vocabulary Test measures receptive vocabulary and is administered to children aged 30 months and older. Due to differences in items from the English version, the Peabody test was normalized using Hispanic norms.

The non-cognitive assessment used in this study is the Child Behavior Check List, which measures socio-emotional development. This parent-completed checklist consists of 99 items that evaluate a range of problem behaviors related to socio-emotional well-being. Higher scores indicate poorer socio-emotional outcomes; therefore, I have reversed the scores to interpret them as negative indicators, reflecting worse outcomes. The CBCL includes two versions for different developmental

[6]For instance, wave 2012 has a sample of children who were born up to 2011 while 2017 interviewed children born up to 2016. Therefore, these two waves are representative of the children in Chile at the time of the release of the wave.

[7]The use of self-reported data for sensitive behaviors like smoking, drinking and consumption of recreational drugs is subject to the recall error typical of large scale retrospective surveys. In this context, measurement error only threatens identification if reporting accuracy shifted differentially between earthquake-affected and unaffected municipalities. The raw trends for risky behaviors across placebo cohorts (2006–09 and 2011–16) show no systematic divergence between these groups (see Figure D.4), particularly for the smoking and risky behavior index. This stability suggests that any measurement error is likely classical and non-differential, which would typically bias estimates toward zero. While a differential shift in recall accuracy could theoretically bias results in either direction, the pre-existing parallel trends in reporting provide no evidence of such a correlation with treatment status. Nonetheless, the reliance on retrospective self-reporting remains a limitation

[8]The reason to use these tests is that they are available for all cohorts through the waves.

stages: the first version assesses children from 18 to 72 months, while the second version is for those 73 months and older. Both sets of scores are normalized using Hispanic norms.

To facilitate comparison across age groups, test scores are transformed into z-scores (mean 0, SD 1) within each age bracket. This allows for the estimation of effects in terms of standard deviation shifts relative to the age-specific mean.

It is important to clarify the composition of the treatment and control groups across the two analysis periods. For the short-term impact, I compare children who were exposed to the earthquake (born 2006–2010, measured in the 2012 ELPI wave) against children born after the earthquake (born 2011–2015, measured in the 2017 ELPI wave). This allows for a comparison of children at similar developmental ages (early childhood). In contrast, the medium-term impact is estimated using only the 2017 ELPI wave. In this specification, the affected cohorts (then aged 7–11) are compared to the post-earthquake control cohorts (aged 2–6) measured at the same point in time. Table C.2 in the Appendix provides a summary of these cohort and survey wave assignments.

The second source of data is the CASEN Earthquake Panel 2009-10, a nationally representative household panel survey conducted initially in 2009 and again in April 2010 in response to the earthquake. This survey includes socio-demographic, employment, and income information, as well as questions regarding the aftermath of the earthquake, such as stress levels and household damages. It is particularly valuable for measuring the short-term income effects, in contrast to the ELPI, which collects income data two years after the event. The third source is the Vital Records data, which encompasses all births registered in Chile between 2008 and 2015. This data enables to examine outcomes at birth, migration patterns, and fertility decisions, while also providing robustness checks for some results derived from the ELPI survey.

Finally, I use the Peak Ground Acceleration (PGA) that represents the maximum ground acceleration recorded at a specific location and is highly correlated with structural damage. This is a continuous measure that captures the non-linearity nature of the intensity of the earthquake. I perform an analysis using PGA as the exposure variable as a robustness check of the main results. This data is obtained from the shake maps provided by the United States Geological Survey (USGS).

3.2 Descriptive Statistics

I start by comparing the earthquake and non-earthquake municipalities. Table 1 presents the balance test, showing that the two groups are remarkably similar across most dimensions, particularly regarding parental schooling and household income. For example, the difference in maternal education is only 0.14 years, and the gap in household income is statistically and economically insignificant.

While I observe a slightly higher probability of a father being present and a small age gap among mothers in earthquake-affected areas (≈ 0.9 years), these differences are minor in magnitude. To address any potential concerns regarding these baseline variations, I include the full set of maternal and household characteristics, including age, education, and family structure, as covariates in all regression models. This ensures that the estimated impacts on child development are not confounded

by these initial differences, but rather reflect the causal impact of the 2010 earthquake. Moreover, using these household and mother characteristics as dependent variable in 1, the Table C.6 shows that the correlations between these characteristics and the earthquake are not statistically significant and small in magnitude to contaminate the results.

Table 1: Descriptive Statistics

	(1) Non-Earthquake	(2) Earthquake	(3) Difference
Children	2.010	2.073	0.063
	(0.970)	(0.964)	(0.041)
HH Size	4.486	4.602	0.115
	(1.643)	(1.557)	(0.122)
Father Present	0.575	0.626	0.052**
	(0.494)	(0.484)	(0.024)
Mothers' Age	33.097	34.040	0.942***
	(7.509)	(7.619)	(0.237)
Mothers' Schooling	12.326	12.465	0.139
	(2.842)	(3.112)	(0.186)
Fathers' Schooling	12.155	12.398	0.243
	(2.991)	(3.320)	(0.276)
Birth Order	1.719	1.745	0.026
	(0.881)	(0.873)	(0.032)
Income (CLP 2017)	244,853.750	257,220.828	12,367.076
	(211,518.109)	(313,070.250)	(19,368.051)
Observations	3,922	11,905	15,827

Notes: ELPI uses sampling weight at national level. Fathers' schooling available only for present fathers. * p < 0.1, ** p < 0.05, *** p < 0.01.

To complement the main analysis and ensure the external validity of the ELPI sample and the validity of the other sources too, I compare its descriptive statistics against the CASEN earthquake panel and Vital Records data in Table C.4. While ELPI mothers tend to be slightly older and are less likely to be married or cohabiting than those in the other datasets, the samples remain remarkably consistent regarding maternal education and family size. To further establish comparability, I specifically benchmark the ELPI mothers, those with children alive at the time of the shock, against the corresponding demographic subgroup in the CASEN Earthquake Panel.

As shown in Table C.9, while some differences between the two sources reach statistical significance, they are small in magnitude. Because p-values identify the presence of a difference rather than its scale, I complement with the Standardized Mean Difference. This approach ensures that we assess the practical magnitude of imbalances rather than mere statistical noise driven by large sample power[9]. These differences prove to be small, half of them fall below the 0.1 threshold and

[9]The Standardized Mean Difference, or Cohen's d, is calculated as the difference in means between the two groups divided by the pooled standard deviation: $d = \frac{\bar{x}_1 - \bar{x}_2}{s_{pooled}}$. Unlike t-tests or p-values, the SMD provides a measure of the magnitude of the imbalance that is independent of sample size. This is particularly relevant when comparing large datasets, where even minor differences can reach statistical significance. Following Austin 2011 and Cohen 1988, an SMD of less than 0.1 is generally considered a negligible difference, while 0.2 represents a small effect size and accepted especially across different data sources.

the other half below 0.2 that is considered small effect size. This confirms that the samples were similar prior to the earthquake and suggests that any statistically significant gaps are small to introduce meaningful bias into the results. To ensure that any baseline differences do not confound the estimated impacts on child development, I include the full suite of maternal and household characteristics, including age, education, and family structure, as covariates in all primary regression specifications. This ensures that the results reflect the causal impact of the earthquake rather than pre-existing regional disparities.

3.3 Empirical Strategy

The identification strategy relies on comparing outcomes of children of the same age in affected and non-affected municipalities. To this end, I employ a difference-in-differences (DiD) approach that takes advantage of the plausibly exogenous seismic intensity across the territory. I use two sources of identification commonly used in the literature (Akresh et al. 2012, Weldeegzie 2017, Rosales-Rueda 2018, Aguilar and Vicarelli 2022). First, I exploit geographic variation by comparing children in municipalities that experienced greater earthquake impacts with those in less affected areas. Second, I use cohort variation by comparing children conceived before the earthquake, from in utero to four years old at the moment of the earthquake, with those conceived after the event. Consequently, I estimate:

$$Y_{imtw} = \alpha + \beta \, \text{Affected}_{it} \times Earthquake_m + X_{imt} + \psi_m + \theta_t + \eta_w + e_{imtw} \tag{1}$$

where i indexes children, m municipality, t cohort of birth, and w survey wave (2012 and 2017). Y is the outcome for the children. Affected_{it} is a dummy variable indicating if the child i belonging to the cohort born in year t was between in utero and up to four years old during the earthquake. $Earthquake_m$ is a dummy variable that takes the value one if the municipality m was impacted by the earthquake. Following Astroza et al. 2010, I classify as the earthquake all municipalities in the Valparaíso region, the Metropolitan region, the O'Higgins region, the Maule region, the Biobío region and the Araucanía region[10]. X_{imt} are household and child characteristics. ψ_m are municipality fixed effects that control for variables that are common to all children in the same municipality. θ_t are birth cohort fixed effects that control for any common shock to children from the same cohort. η_w is a survey fixed effect that controls for variables that are common to all children surveyed in a specific wave[11]. Finally, e_{ijtw} is a random error term. To allow for possible spatial and time correlation, cluster standard errors are calculated at the municipality level.

One potential concern with this identification strategy is the presence of unobserved, time-varying

[10] In Table C.26, I depart from the official classification to define impacted municipalities based on their Peak Ground Acceleration (PGA) intensities. By testing various cutoffs at the 25th, 50th, and 75th percentiles of the PGA distribution, I find that the results remain highly consistent with the estimates obtained using the official government classification.

[11] I also run another specification when I pool the development outcomes from the 2012 and 2017 waves for the analysis in the short-term, where I interact X_{imt} with η_w. The results are similar to the main ones and are available upon request.

characteristics at the municipality level that could be correlated with both earthquake intensity and child development. While municipality fixed effects (ψ_m) account for time-invariant differences, they do not capture divergent developmental trajectories between affected and unaffected areas. To address this, I perform several checks. First, I include municipality-specific linear trends to control for the possibility that child outcomes were already following stable yet differing paths across municipalities. This strategy strengthens the identification by allowing for these differential trends; it rests on the premise that, absent the earthquake, the trajectories of children in affected areas would have paralleled those of their peers in unaffected municipalities. Second, I conduct placebo tests on cohorts born after the shock. As I will discuss in Section 3.4, these estimates suggest that unobserved time-varying heterogeneity is unlikely to be driving the results, as the outcomes for these post-shock cohorts do not systematically vary with earthquake intensity.

The parameter β is interpreted as an intention-to-treat parameter, in particular, it measures the average effect on children in municipalities exposed to the earthquake on human capital outcomes[12]. Since to control for municipality fixed effects, I compare development outcomes with children who were conceived after the earthquake, this group might not be a perfect control group since they may have been indirectly affected by the earthquake through mothers' stress or income dynamics[13]. If this were the case, I would expect that the earthquake affected more strongly the children who were directly affected by the earthquake, those in utero or already born, and to a lesser extent those conceived after. For this reason, I would interpret β as a lower bound of the true effect[14].

To estimate a causal effect, the main identification assumption is that, in the absence of the earthquake, outcomes for affected and unaffected children would have followed similar trends. This relies on two key assumptions. First, children born after the earthquake must serve as a valid counterfactual to control for unobservable differences across municipalities. This requires that unobserved heterogeneity at the local level remains consistent across birth cohorts and is adequately captured by year-of-birth fixed effects. Second, any cohort-related differences in unobserved characteristics must be comparable between affected and unaffected municipalities, which are accounted for through municipality fixed effects.

3.4 Validity of the Identification Strategy

To assess the validity of the estimates, I test several potential threats to the identification assumption. In the following paragraphs, I present the evidence that supports the validity of the estimates. Specifically, both the parallel trend assumption (validated in the placebos cohort using an event cohort study) and placebo estimates show that the children who were born after the earthquake serve as a good control group for children who were from in utero up to four years old during the

[12] In the specification, I have three control groups: the children living in the least affected municipalities and the children who were conceived after the earthquake in affected and unaffected municipalities.

[13] Although I do not find a significant effect on income two years after the shock in Column 1 in the Table 7.

[14] In Section B.2, I evaluate the validity of using the 2011 cohort as a baseline. I argue that this group remained largely insulated from the acute disruptions experienced by the 2006-10 cohorts, making them a suitable control group for the analysis.

earthquake.

Parallel Trends. The identification of the causal effect relies on the assumption that pre-treatment trends in cognitive and non-cognitive outcomes for the placebo cohorts (2011–2015) are balanced across earthquake-affected and non-affected municipalities. To test this, I estimate an event cohort study specification by interacting the birth cohort with the earthquake treatment indicator[15]. Figures 2 and 3 present the results for short- and medium-term outcomes, respectively. In these figures, cohorts are ordered such that the placebo (younger) cohorts appear first, providing a visual test of the parallel trend assumption to the earthquake exposed cohorts (2006–2010). The 2011 cohort serves as the excluded reference category.

For the short-term outcomes (Figure 2), the estimates generally support the parallel trends assumption. For the Peabody test the coefficients for the 2012–2014 cohorts are statistically indistinguishable from zero. An exception occurs in the 2015 cohort, which exhibits a significant negative coefficient; I will demonstrate below that this specific anomaly does not constitute a violation of the parallel trend assumption. Regarding earthquake exposure, the 2006–2010 cohorts show a consistent negative impact on Peabody scores and significant for some cohorts, reinforcing the main DiD estimates. Results for Battelle and CBCL are less definitive in the cohorts 2006-10, with coefficients being largely non-significant.

Medium-term results (Figure 3) similarly support parallel trends for the 2012–2014 Peabody cohorts and the 2012 CBCL cohort[16]. Two caveats warrant mention. First, the 2015 Peabody coefficient remains negative and significant. Second, the CBCL placebo window is limited by age requirements; because the CBCL for older children requires children to be at least five years old, only the 2011 cohort and one-third of the 2012 cohort were eligible for testing. The estimate of the 2012 cohort coefficient is noisy and it supports the assumption by being not significant, however, the estimate may not be fully representative of the entire cohort. Nevertheless, the impact on exposed children (2006–2010) remains clear, particularly for Peabody and CBCL, where coefficients are consistently negative and significant in the case of Peabody.

I also address the 2015 Peabody anomaly by providing evidence that the significant coefficient for the 2015 cohort is a measurement artifact rather than a failure of the identification strategy. As the youngest cohort (age 2.5 at test), the 2015 group represents a psychometric boundary case. Standardized data (Tables B.2 and B.1) reveals that the 2015 cohort exhibits a clear psychometric floor effect, where the test's inability to capture low-end variation results in a mechanical mean inflation (105.59 vs. 104.58 in 2014) and compressed variance (SD of 12.84 vs. ≈ 18.5). By effectively censoring the lower tail of the distribution, evidenced by a minimum score of 80 compared to 66 in older cohorts, the mean is forced upward, creating a paradoxical mean inversion that masks true developmental trends. Furthermore, as the smallest sample in the study ($N = 512$), this group is likely subject to selection bias, as only the oldest children within that birth year met the age 2.5 eligibility requirement. These patterns suggest the Peabody lacks the resolution to distinguish

[15] I also present the raw trends for each test in the appendix that complement the event study analysis in figures D.2 and D.3

[16] As a reminder, the placebo cohorts are the same for short- and medium term Peabody analysis.

developmental hierarchies at this lower age bound, rendering the 2015 cohort a statistical outlier driven by measurement constraints rather than cognitive ability.

Critically, the parallel trend assumption holds for the placebo cohorts immediately following the earthquake (2011–2014). A joint F-test for these years is non-significant ($p = 0.25$), confirming that treated and control groups followed identical trajectories. The results are also structurally stable: excluding the 2015 cohort renders the DiD estimates invariant to the inclusion of municipality-specific linear trends. While the inclusion of linear trends doubles the treatment effect when the 2015 cohort is included (tables 2 and 3), indicating that the model is attempting to correct for this spurious boundary glitch, the estimates remain stable in magnitude when the 2015 cohort is removed (see Table C.22). This confirms that the primary findings are driven by the 2010 seismic shock rather than pre-existing regional trajectories. (See Appendix Section B.1 for further discussion).

Finally, I find no significant heterogeneity by age across the cognitive and non-cognitive tests, likely due to the increased standard errors associated with subsample cohort analysis.

Placebo estimates. One potential concern is that the estimated effects may be confounded by underlying time trends or omitted variables. To address this issue, I conduct a placebo test by constructing a fake earthquake using only the sample of children conceived after the actual 2010 earthquake. In this analysis, I replicate the same classification of municipalities impacted by the 2010 earthquake and divide the post-earthquake sample into two equal groups. The older half is designated as the treated cohort, analogous to the main specification, while the younger half serves as the control group, simulating exposure to a fictitious shock. If the results were driven by secular trends or unobserved confounders, one would expect to find statistically significant coefficients in this placebo setting.

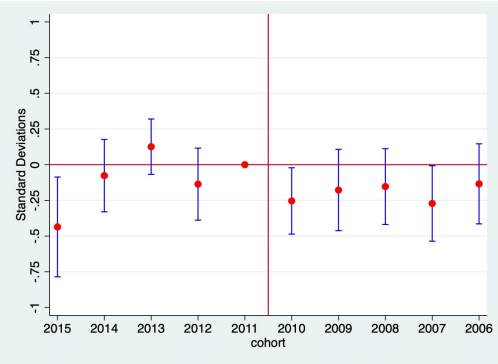
The results, presented in Table C.14, show that none of the estimates are statistically significant, suggesting that the main findings are unlikely to be the result of time trends or omitted variable bias[17]. It is also important to note that for the CBCL2 measure, only the 2011 birth cohort and a small portion of the 2012 cohort are included in the sample of children born after the earthquake.[18] To account for this, I divide this limited subsample into two equal groups.

Falsification tests. I run two falsifications tests where I exclude municipalities that have been classified as affected by the earthquake, and use those non-affected, the northern and southern regions of the country. Since these regions are classified as unaffected, I should not find any significant difference between them. In the first falsification test, I regress Equation 1 using only these unaffected regions and the PGA intensity measure in the municipalities in these regions. As anticipated, Table C.16 shows no significant results for any of the cognitive and non-cognitive tests. In the second falsification test, I define affected by the earthquake the northern regions and non-affected, the southern ones. I create a binary treatment variable that takes the value of one if they are in northern regions and zero elsewhere. Table C.17 presents imprecise and noisy estimates.

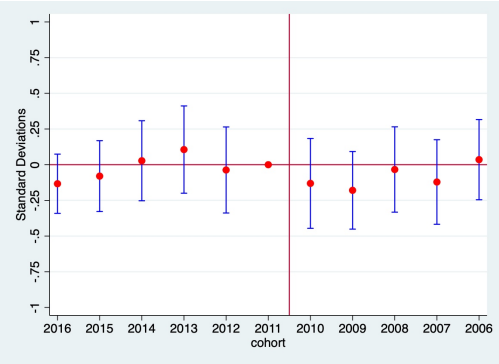
[17]Note that for the Peabody test, the group of children born after the earthquake serves as the control group for both the short-term and medium-term measures of test performance.

[18]This is because the CBCL2 is administered starting at 73 months of age, and most children born after the earthquake had not yet reached the eligible age at the time of measurement.

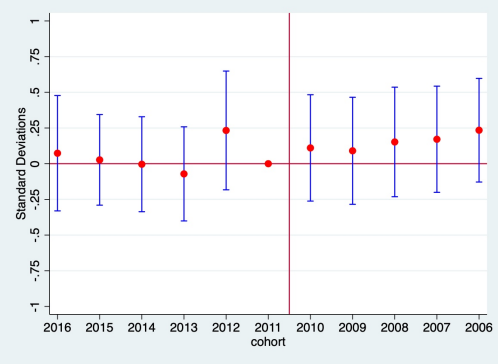
Figure 2: Event Cohort Study: Short-Term Development



(a) Peabody



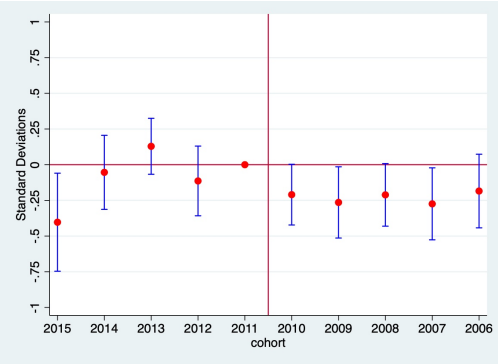
(b) Battelle



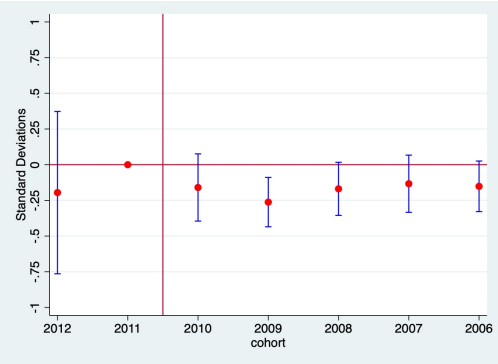
(c) CBCL

Notes: This figure plots the coefficients and 95% confidence intervals from an event-study specification interacting birth cohorts with the earthquake treatment indicator. The 2011 birth cohort serves as the excluded reference category. Placebo cohorts (2011–2015) appear first to provide a visual test of the parallel trends assumption, followed by earthquake-exposed cohorts (2006–2010). Dependent variables are standardized test scores: Battelle, Peabody and CBCL measured two years later for the affected cohorts. Placebo cohorts were measured seven years later. All regressions include controls for child gender, age in months at test time, birth order, mother's age and schooling, presence of father, household size, and prior mental health conditions of parents/relatives. Estimates use national sampling weights. Standard errors are clustered at the municipal level (145 clusters).

Figure 3: Event Cohort Study: Medium-Term Development



(a) Peabody



(b) CBCL

Notes: This figure plots the coefficients and 95% confidence intervals from an event-study specification interacting birth cohorts with the earthquake treatment indicator. The 2011 birth cohort serves as the excluded reference category. Placebo cohorts (2011–2015) appear first to provide a visual test of the parallel trends assumption, followed by earthquake-exposed cohorts (2006–2010). Dependent variables are standardized test scores: Peabody and CBCL. Affected children along with placebo cohorts were measured seven years later. All regressions include controls for child gender, age in months at test time, birth order, mother's age and schooling, presence of father, household size, and prior mental health conditions of parents/relatives. Estimates use national sampling weights. Standard errors are clustered at the municipal level (145 clusters).

The results provide no evidence of significant differences among these groups of regions in terms of cognitive and non-cognitive development.

Migration. Another potential concern is that families migrated away from areas affected by the earthquake to non-affected ones. If this were to be the case, the main parameter would be underestimated. To assess this possibility, first, I calculate migration between ELPI waves 2010 and 2012. I found a small 0.78% migration in the sample, with migration from affected to unaffected areas was 0.02%. Additionally, I use Vital Records from all births in Chile and the set of mothers who gave birth at least twice in the sample, with one child before and another one after the earthquake. I found that only 1.32% migrate from affected to unaffected places. In general, most migration happens among affected municipalities mainly because 80% of the population lives in the affected zone. To formalize the previous analysis, Table C.18 presents a model of mother fixed effects[19]. I find significant effects but they are too small in size to explain the results. Columns 1 and 2 show no effect on the probability to migrate to another municipality or region in the most affected municipalities/regions respectively. Columns 3 and 4 show small effects on the probability to migrate from a non-affected (NA) municipality to an affected one (A) and from a NA to an A one. To summarize, the migrations effects are too small to bias the main estimates.

Endogenous fertility. It may be the case that families delay birth decisions. In the case that mothers from poorer (wealthier) backgrounds delay it, this can positively (negatively) select the

[19]The model exploits month/year and municipality variation on the probability of migration. The choice of mother fixed effects instead of father fixed effects is because the only identifier in the Vital Records is mother ID.

control group of children conceived after the earthquake biasing towards zero (negative effects) the estimates. Table C.19 shows suggestive evidence of no effect. Column 1 exploits time and municipal geographic variation using the information of all births in Chile from the Vital Birth Records [20], I find that the number of children born does not vary significantly in the most affected municipalities after the earthquake. Column 2 takes advantage of the question from the ELPI survey "whether the child was planned" and the specification from Equation 1. I find that pregnancy decisions in the most affected municipalities were not differently planned after the earthquake with respect to non-affected municipalities.

To further address the potential threat of endogenous fertility, I compare observable characteristics of mothers in affected and unaffected municipalities before and after the earthquake. For example, if mothers who conceived after the earthquake come from better socio-economic backgrounds, this could lead to an underestimation of the true effect. To investigate this concern, I use the Equation 1 and household and parents observable characteristic as dependent variables. The results presented in Table C.6 show a small and statistically not significant relationship between parents and household characteristics and the earthquake[21].

Overall, these findings provide suggestive evidence that pregnancy decisions were uncorrelated to the earthquake occurrence and that endogenous fertility is unlikely in this setting.

Selective mortality. A critical concern in studies of early-life shocks is selective mortality, which can lead to a positively selected sample of survivors, in other words, a stronger treatment group. If the earthquake disproportionately affected the survival of the most vulnerable fetuses, particularly males, who are biologically more sensitive to maternal stress (Trivers and Willard 1973), the estimates would be biased toward zero. To objectively assess this risk, I follow the methodology of Karbownik and Wray 2019 and examine the impact of the earthquake on the gender ratio at birth. Using aggregated Vital Records and ELPI survey data at the municipality/month-by-year level, Columns 3 and 4 of Table C.19 show no evidence of a shift in the gender ratio in treated municipalities.

This lack of impact on the gender ratio suggests that the earthquake did not induce significant selective fetal loss. Combined with evidence that casualties among children under nine years old were negligible (Lastra et al. 2012), these results support the assumption that the treatment and control groups remain comparable in their underlying characteristics[22]. Therefore, the main estimates likely reflect the average effect of the earthquake on the total population, rather than an artifact of compositional changes due to differential survivor bias.

Attrition rate. A potential concern in the analysis is that sample attrition may be systematically

[20] Instead of birth cohort and geographical variation as in ELPI data.

[21] To further provide evidence, I use the vital records data using the universe of births (around 739,000 births) by comparing mother's characteristics who conceived before and after the shock using 1. I find significant results that are precisely estimated but small enough in magnitude that lack from economic meaning. These results are shown in Table C.8 in the appendix.

[22] Lastra et al. 2012 document that the number of casualties of children below nine years old was 2.09 deaths by 100,000 population, similar to the ones in all age range of people below 50 years old and smaller than those older than 50 years old.

correlated with the treatment variable, which could bias the estimated effects. In order to control for this problem, in the paper I use sampling weights to account for attrition and maintain national representativeness. However, to further investigate this possibility, I use data from the 2010 and 2012 waves of the ELPI survey. Specifically, I regress an indicator variable equal to one if the child was interviewed in either the 2010 or 2012 wave but is missing in the 2017 wave on the measure of earthquake exposure, using Equation 1[23].

The results, presented in Table C.20, show that attrition is, on average, 6 percentage points higher for children exposed to the tremor in impacted municipalities. I hypothesize that this higher attrition in impacted areas may attenuate the estimated effect of the earthquake on children, as those more severely affected are also more likely to drop out of the sample. Consequently, the estimates presented should be interpreted as lower bounds of the true impact.

Although Alderman et al. 2001 suggests that this level of attrition is relatively low for surveys conducted in developing countries, I implement bounding techniques as proposed by Lee 2009 and Kling et al. 2007[24]. These bounds are calculated without sampling weights, as such weights are unavailable for individuals missing from the 2017 wave.

To do this analysis, first, I show that the estimates without sampling weights in Table C.32 are quantitatively similar to the ones with weight in Tables 3 and 2, indicating that the main findings are not driven by the weighting scheme. Next, using the sample without sampling weights, Table C.21 presents the results from the Lee and the Kling and Liebman bounding approaches. The findings for the Peabody and CBCL2 outcomes remain robust under the Lee bounds and under the ± 0.1 SDs bounding scenarios. However, when outcome values for attritors are replaced using ± 0.25 SDs, the estimates are no longer significant although they continue having the correct signs. It is important to note that this scenario implies a 0.5 SD difference in outcomes between attritors in the treatment and control groups—an extreme and likely unrealistic assumption. Moreover, since attrition is higher in the most severely affected municipalities—and since it is more likely that these children were more negatively impacted by the earthquake—the most relevant bounds would be the upper bounds, which consistently yield significant negative effects. Nonetheless, adopting a conservative interpretation, the results for the Peabody and CBCL2 measures remain robust under both the Lee bounds and the ± 0.1 SD sensitivity analyses.

[23]While the main analysis utilizes the 2012 and 2017 waves, the 2010 ELPI wave is used exclusively to test for differential attrition in the immediate aftermath of the disaster. This pre-disaster baseline is essential to verify that the earthquake did not cause systematic sample selection (e.g., via migration or mortality) between the event and the first follow-up in 2012, which would bias the subsequent estimates.

[24]Lee's (2009) approach addresses attrition bias by constructing bounds that trim the sample based on the differential attrition rates across groups. This method assumes that all missing observations come from either the top or the bottom of the outcome distribution, depending on the direction of attrition. Alternatively, a more flexible and informative method is the sensitivity bounds approach proposed by Kling and Liebman (2004), which evaluates how results vary under different assumptions about the extent of selection bias. In the context of the earthquake, for example, one can test the robustness of the estimates by assuming that children who were not re-interviewed would have scored 5%, 10%, or even 25% higher or lower than those who were successfully re-interviewed. This method offers a structured framework to assess the sensitivity of the findings to potential non-random attrition.

4 Results

4.1 Short-Term Results

I estimate the effect of the earthquake on cognitive and non-cognitive measures two years after the earthquake. I analyze a cognitive test, a non-cognitive and a blend of the two. The latter one is the Battelle test, which measures a composite indicator of different abilities such as communication, social-emotional, adaptive, motor, and cognitive skills. Column 1 shows the specification with only municipality fixed effects. In Column 2 adds birth cohort fixed effects. Column 3, the preferred specification, adds controls for child, mother, and household characteristics. In Column 4, as a robustness check, I incorporate municipality linear trends to address the possibility of stable yet differing trends in child outcomes across municipalities.

Column 3 in Panel A of Table 2 shows a 0.08 SDs reduction in the Battelle test but not significant. It increases its negative effect to a reduction of 0.23 SDs and becomes significant after adding municipality linear trends.

Next, I examine the effect on Peabody test scores in the short-term. Column 3 in Panel B of Table 2 indicates that children exposed to the earthquake reduced their Peabody test scores in an average of 0.14 SDs being this effect significant at standard levels. However, as shown in Table C.33 in the appendix, this short-term cognitive effect is fragile since it does not survive Romano-Wolf multiple hypothesis testing correction (adjusted p-value of 0.12). As with other cognitive measures, incorporating municipality linear trends increases the effect's absolute magnitude.

Finally, I focus on the non-cognitive estimates in the short-term. Table 2 presents the results of the CBCL scores on Panel C. In this case, I found non-significant effects across all specifications including the one with municipality linear trends.

4.2 Medium-Term Results

In this section, I present the estimates of the earthquake's effect on cognitive and non-cognitive development seven years later in Table 3. The dependent variables are the normalized Peabody test scores in Panel A and the CBCL2 scores in Panel B.

The results in Panel A from Columns 1 to 3 demonstrate a consistent negative effect on Peabody test scores for children who experienced the adverse consequences of the earthquake. In Column 3 indicates that children in the most earthquake municipalities reduced their test scores in 0.17 SDs. This effect is economically meaningful; according to D. Evans and Yuan 2019, a reduction of 0.17 standard deviations in literacy skills corresponds to an average loss of 0.8 years of schooling for children in the most affected municipalities at the moment that these children are measured. In terms of magnitude, the estimate is smaller to the findings of Andrabi et al. 2021, who report that children in the epicenter area reduced their test scores in 0.27 SDs compared to children who lived 30 kms away due to the 2006 Pakistan earthquake[25]. Additionally, Column 4 shows a 0.28 SDs

[25]The finding of a 0.17 SD reduction in cognitive scores is remarkably consistent with the broader literature on early-life shocks in Latin America. As summarized in Table C.3, the results align closely with the impacts found for

Table 2: The Impact of the Earthquake on Cognitive and Non-Cognitive Development in the Short-Term

	(1)	(2)	(3)	(4)
Panel A: Battelle				
Affected*Earthquake	-0.0978	-0.108	-0.0813	-0.232*
	(0.0938)	(0.0926)	(0.0944)	(0.137)
Observations	13,070	13,070	13,070	13,070
$Adj - R^2$	0.0562	0.185	0.224	0.260
Panel B: Peabody				
Affected*Earthquake	-0.148	-0.158*	-0.141*	-0.292***
	(0.0917)	(0.0890)	(0.0855)	(0.108)
Observations	14,205	14,205	14,205	14,205
$Adj - R^2$	0.0660	0.0928	0.142	0.163
Panel C: CBCL1				
Affected*Earthquake	0.0978	0.0939	0.126	0.0428
	(0.104)	(0.103)	(0.100)	(0.144)
Observations	11,654	11,654	11,654	11,654
$Adj - R^2$	0.0733	0.0781	0.137	0.168
Municipality FE	✓	✓	✓	✓
Birth Cohort FE	×	✓	✓	✓
Child-Mother-HH control	×	×	✓	✓
Municipality linear trends	×	×	×	✓

*Notes: Each column represents a separate regression. The dependent variable is the test score measured in standard deviations. Peabody is the Peabody Test Score, Battelle is the Battelle Test Score and CBCL1 is the Child Behavior Check List measure. The scores for affected cohorts (2006-2010) are measured two years later and the placebo cohorts (2011-2015) are measured 7 years later. Observations are at the individual/municipality level. Column 1 includes an indicator variable $Affected_{it}$ that captures if the child was alive or in utero and the moment of the earthquake. Standard errors clustered at the municipal level appear in parentheses—number of clusters: 145. The estimates use sampling weights representative at the national level. Child/Mother/HH characteristics include gender, age at the moment of the test (in months), birth order, mothers' age and schooling, father present, number member of the household, prior mental health condition for mother, father and relative. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.*

Table 3: The Impact of the Earthquake on Cognitive and Non-Cognitive Development in the Medium-Term

	(1)	(2)	(3)	(4)
Panel A: Peabody				
Affected*Earthquake	-0.180**	-0.190**	-0.174**	-0.280***
	(0.0752)	(0.0732)	(0.0686)	(0.0866)
Observations	14,069	14,069	14,069	14,069
$Adj - R^2$	0.115	0.132	0.185	0.197
Panel B: CBCL2				
Affected*Earthquake	-0.123*	-0.126*	-0.129*	-0.220**
	(0.0759)	(0.0762)	(0.0763)	(0.105)
Observations	11,568	11,568	11,568	11,568
$Adj - R^2$	0.0659	0.0657	0.107	0.118
Municipality FE	✓	✓	✓	✓
Birth Cohort FE	×	✓	✓	✓
Child-Mother-HH control	×	×	✓	✓
Municipality linear trends	×	×	×	✓

*Notes: Each column represents a separate regression. The dependent variable is the test score measured in standard deviations. Peabody is the Peabody Test Score and CBCL is the Child Behavior Check List measure. These scores are measured seven years later. Observations are at the individual/municipality level. Column 1 includes an indicator variable $Affected_{it}$ that captures if the child was alive or in utero and the moment of the earthquake. Standard errors clustered at the municipal level appear in parentheses—number of clusters: 145. The estimates use sampling weights representative at the national level. Child/Mother/HH characteristics include gender, age at the moment of the test (in months), birth order, mothers' age and schooling, father present, number member of the household, prior mental health condition for mother, father and relative. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.*

reduction when adding municipality linear trends which provides robustness to the main result[26].

Panel B of the Table 3 shows the effect on non-cognitive scores on CBCL2. All the estimated effects from Columns 1 to 4 are negative and statistically significant. Column 3 reveals that children in the most impacted municipalities reduced their non-cognitive test scores in 0.13 SDs. Although the literature documenting the impact of negative shocks during in utero and early childhood on non-cognitive development is limited, the results are in line with Chang et al. 2022 who identify a significant reduction 15 years later on children in utero affected by rainfalls but not in middle years. Also, other papers have found negative effects of famines on personality (Stein et al. 2009, Neugebauer et al. 1999, Canavan et al. 2013). The addition of municipality linear trends in Column 4 increase the negative effect as with the Peabody test.

A notable puzzle in these results is the difference in timing across developmental domains. While the estimates suggest early disruption in receptive vocabulary (Peabody), the medium-term effects are considerably more robust and persist, the effects on non-cognitive development (CBCL) and the composite Battelle index are less clear in the short-term. Specifically, the CBCL exhibits a statistically significant negative effect only seven years later, while the Battelle index, which mixes cognitive and non-cognitive elements, shows no significant effect early on.

This pattern suggests that while the evidence points to an early disruption in foundational cognitive skills early in life, other developmental consequences remained latent during the first years after exposure. This interpretation is consistent with the notion of latent disadvantages. As argued by Currie and Almond 2011 and Conti et al. 2018, early-life shocks can alter children's developmental trajectories without immediately affecting observable outcomes. As developmental demands become more complex, particularly with the transition into school, deficits that were previously difficult to detect may become increasingly apparent.

One possible explanation for the absence of an early effect on the Battelle index is that it combines cognitive and non-cognitive domains into a single composite measure. If the earthquake initially affected primarily cognitive skills, while non-cognitive deficits emerged only later, the composite index may have been less sensitive to these early disruptions. The medium-term results, together with the findings of Chang et al. 2022, suggest that the developmental consequences of severe early-life shocks may not be fully observable during the first years of life, but instead become more evident as children age and developmental demands increase.

floods in Mexico (0.19 SD, Aguilar and Vicarelli 2022) and conflict in Colombia (0.18 SD, Duque 2017). While the estimates are larger than those found for rainfall shocks in Ecuador (0.1 SD, Rosales-Rueda 2018), they remain more conservative than the severe long-term impacts observed following the 1970 Peruvian earthquake, which resulted in a 0.5 year reduction in total schooling (Caruso and Miller 2015). The fact that a 0.17 SD drop in Chile represents a 0.8 year schooling equivalent—surpassing the effects of some historical Peruvian disasters—underscores the significant human capital cost of the 2010 tremor, even within a high-income regional context.

[26]To address potential confounding effects from the tsunami, I conducted a triple-difference (DDD) analysis by interacting the *Affected* × *Earthquake* treatment term with a dummy variable for coastal municipalities and the lower-order interactions. As shown in Table C.25 of the Appendix, the triple interaction coefficients are statistically insignificant across all outcomes and small in magnitude. This suggests that coastal municipalities did not follow a systematically different recovery or developmental trajectory relative to inland affected areas, supporting the conclusion that the main effects are driven by the earthquake's broader impact rather than localized tsunami damage.

4.3 Robustness Checks

Peak Ground Acceleration and Distance to the Epicenter as Exposure Variable. I use Peak Ground Acceleration (PGA) as a measure of the intensity of the earthquake. This is a continuous measure that captures the non-linearity nature of the intensity of the earthquake. PGA is measured in $meters/seconds^2$ (m/s^2) and has been widely used in earthquake engineering since it measures the maximum acceleration or the local ground shaking recorded at a particular location during an earthquake. This measure is highly correlated with structural damage. The results from these estimates using PGA as exposure or intensity variable are shown in Table C.23. Overall, the estimates are similar in sign and significance to the main estimates from Tables 3 and 2. Additionally, I use distance to the epicenter of the earthquake in Table C.24 and the results are consistent with the main results in the short and medium-term.

Classification of Impacted Municipalities, Regional Cluster and Regional Fixed Effects. A potential concern in the econometric setup is whether fixed effects and clustering should be defined at the regional level, given the earthquake's broad geographic reach. However, I prioritize municipality level fixed effects for three primary reasons. First, seismic intensity measured by PGA exhibits substantial within region heterogeneity; for instance, in the Biobío Region, PGA values range from 0.25 m/s^2 to 0.53 m/s^2 (Table C.12). Using region level fixed effects would therefore absorb this economically meaningful variation in exposure. Second, this local variation provides the granularity necessary to capture a dose response relationship, where developmental setbacks scale proportionally with the intensity of the local shock rather than being treated as a uniform regional event. This is further validated by the specifications using continuous PGA, which confirm that the magnitude of the impact follows a monotonic relationship with the physical intensity of the earthquake at the municipal level. Third, the institutional response in Chile is highly decentralized; post-disaster relief, housing subsidies, and school reconstruction were coordinated primarily at the municipal level based on local damage assessments.

To ensure the results are not sensitive to these specifications, I perform robustness checks using region level fixed effects and regional clustering with a Wild Cluster Bootstrap (N=15 regions) to account for the small number of regional clusters. As shown in Appendix Table C.27, the estimates remain remarkably stable in both magnitude and significance, confirming that the findings are not a mechanical byproduct of the level of geographic aggregation. Finally, I test for sensitivity to threshold selection by redefining the earthquake indicator at the 25th, 50th, and 75th percentiles of the PGA distribution. As shown in Table C.26, the results remain robust across these cutoffs. This exercise is critical as it confirms that the findings are not dependent on an arbitrary binary classification or driven by a few extreme outliers; rather, they reflect a consistent relationship between seismic exposure and developmental setbacks across the entire distribution of the shock.

Changing the Control Group. I change the definition of the control group of non-exposed municipalities to the earthquake. First, I define the control group as the nearest municipalities as those adjacent to affected municipalities in both the north and south. Second, I define the control group as the furthest municipalities, where those are located at the northernmost and southernmost

extreme of the country. The drawback of this analysis is the loss of power since I cut off in half the control group. With this caveat in mind, Panels A and B of Table C.28 show that in general the estimates are consistent in magnitude and sign for the Peabody and CBCL2 results, however, losing some significance in some of them.

Dropping the First Cohort Born After the Earthquake (2011). Another concern is that the control group of children conceived after the earthquake may be contaminated by the aftermath of the earthquake. For example, they may have been indirectly affected by the earthquake in terms of mothers' stress or household income dynamics. To shed some light on this issue, I re-estimated the main specification while excluding the group of children born in the following year the earthquake, the 2011 cohort. If the control group were indeed contaminated, I would expect to see a more negative estimated effect. Table C.30 points out that the estimates for all Peabody cognitive development measurements remain similar in magnitude to the main estimates. For non-cognitive measures, the short-term estimates are similar in magnitude but statistically non-significant and in the medium-term, the effect becomes positive and imprecise because excluding the first cohort born after the earthquake reduces cohort of children born after the earthquake to 250 observations. With this caveat in mind, these results provide suggestive evidence that the potential contamination of the control group is not a relevant concern.

Sensitivity to Age-at-Test and Potential Biases. A potential concern in the medium-term analysis is the age discrepancy between the treated group (7-11 years) and the control group (2-6 years) at the time of the 2017 follow-up. While Peabody and CBCL scores are monthly age-normalized using Spanish norms, comparing children across distinct developmental stages introduces potential sources of bias, all of which suggest that my main estimates likely serve as a conservative lower bound:

- **Non-linear Scaling and Maturity Bias:** The Peabody test scales from basic object recognition in early childhood to complex linguistic concepts in middle childhood. If the hurdle to stay at the mean is structurally higher for older children, a persistent -0.17 SD deficit in a 10 year-old may represent a more profound loss in potential than a similar deficit in a 4 year-old. Similarly, for the CBCL, socio-emotional stress manifests differently as children mature. While younger children (5-6 years) often display visible, externalizing behaviors, older children (9-11 years) may experience more quiet internalizing symptoms like social withdrawal or anxiety, which are more easily masked or normalized in parent-reports. This suggests an attenuation bias that may systematically underestimate the cognitive and psychological burden on the older treated group.

- **Environmental Buffering Bias:** The older treated cohorts (7-11) have been exposed to several additional years of formal primary education and social interaction compared to the control group. In a standard human capital framework, these years of formal schooling and socialization should act as a remedial buffer. If the initial shock were a temporary delay, the gap should shrink as children mature and access these institutional resources. In the

estimates, instead, the deficit persists and even slightly widens in Peabody and CBCL tests in the medium-term.

To address these concerns directly, Table C.31 presents local comparisons that minimize the developmental gap. When narrowing the window to cohorts born in 2010 versus 2011, reducing the age gap to approximately 12 months, the estimated impact on Peabody scores strengthens to -0.249 SD. As the window expands (2009-2012), the coefficients remain stable around -0.20 SD. A similar pattern is observable for the CBCL; the effect moves from -0.20 SD in the tight 2010-2011 window to -0.15 SD in the 2008–2012 window, which is consistent with the main effect of -0.13 SD.

These patterns are the most telling part of the analysis. If age-at-test were a confounder, the effects should dissipate as the age gap narrows. Instead, I observe that the effects remain robust and even increase in magnitude when comparing children who are essentially at the same developmental stage today. This confirms that the earthquake shock, rather than age-related confounding, is the primary driver of the results, and that focusing on the most sensitive developmental windows reveals even larger cognitive and socio-emotional scarring.

Sampling Weights and Multiple Hypothesis Testing. I conduct two additional robustness checks. First, I re-estimate the main specification without applying sampling weights as a discussed in the Section 3.4. Since attrition between survey waves can introduce bias, researchers commonly use sampling weights to correct for potential imbalances. While this is a standard approach, estimating the model without weights allows to assess the sensitivity of the results. As shown in Table C.32, the unweighted estimates are quantitatively similar to the baseline results, suggesting that the findings are not driven by the weighting scheme.

Second, I address the issue of multiple hypothesis testing. Given that multiple cognitive and non-cognitive outcomes are examined, the likelihood of false positives increases, raising the chance of finding statistically significant effects purely by chance, even when the null hypothesis is true. To account for this, I apply Romano-Wolf adjusted p-values[27], which are reported in square brackets in Table C.33. The results show that the Peabody and CBCL2 scores remain statistically significant in the medium-term at conventional levels, while the Peabody score in the short-term is more fragile, with a p-value of 0.12.

4.4 Heterogeneity Analysis

Thus far, I have identified the adverse impact of exposure to the earthquake during in utero and early childhood on cognitive and non-cognitive development. However, this effect may not be uniform across all children. In this section, I examine heterogeneous effects in various dimensions, including gender, maternal age and education, paternal presence and education, and income.

[27]To account for the multiple testing of the five developmental outcomes, I employ the Romano-Wolf (rwolf) resampling-based procedure. I prefer this method over Anderson (2008) q-values because it provides more conservative control over the Family-Wise Error Rate (FWER) while explicitly modeling the high degree of correlation between the cognitive and non-cognitive measures. By using the rwolf method, I achieve higher statistical power than traditional Bonferroni or Holm corrections without sacrificing the rigor required for longitudinal analysis.

Gender. I examine the effects by gender. First, in terms of balance, these two samples are fairly balanced similarly as the main sample (see Tables C.5 and 1). Additionally, there is not evidence of selection on observables by gender (table C.7). Second, Columns 1, 2, and 4 in panels A and B of Table 4 indicate that the negative impact on cognitive development is consistently larger for boys than for girls. Specifically, boys in impacted municipalities experienced a reduction in cognitive test scores of 0.25 SD in both the short- and medium-term. In contrast, the effects for girls were much smaller (0.04 to 0.09 SD) and statistically insignificant. Critically, the differences between these gender specific estimates are statistically significant, leading to the conclusion that the adverse effects were driven by the impact on boys, while no significant harm was detected for girls.

In terms of non-cognitive development, Column 3 reveals no significant short-term effects for either boys or girls. By the medium-term (Column 5), the results of the CBCL2 test show a negative and statistically significant effect for boys, whereas no effect is observed for girls. While this pattern suggests that boys may be more adversely affected, the difference between the gender estimates is not statistically significant. Consequently, these results do not provide conclusive evidence of a differential impact between boys and girls.

In summary, the differential results on cognitive development, specifically more negative for boys than girls, align with other papers that explode exposure to earthquakes as negative shocks in early childhood (Bustelo et al. 2012, Thamarapani 2017) or other types of negative shocks in these stages (Weldeegzie 2017, Chang et al. 2022, Bimardhika and Moorena 2025).

Other Characteristics. I explore several other dimensions that traditionally determine a household's capacity to buffer external shocks. Mother's age and the presence of a father at home serve as proxies for household maturity and the availability of parental time and emotional support, both of which are critical for stabilizing a child's environment following a disaster. Similarly, mothers' and father's education and per capita income provide a more complete picture of the family's resource ceiling and their ability to finance remedial investments, such as better nutrition or private healthcare. However, despite these theoretical grounds for variation, the heterogeneous analysis in Tables C.34, C.35, C.36, C.37 and C.38[28] yields no statistically significant results across any of these dimensions[29]. The negative impact of the earthquake on child development appears to be remarkably uniform, suggesting that the scale of the 2010 disaster may have overwhelmed the protective private resources typically available to higher SES or more stable households. The only dimension where a subtle effect seems to show up is mothers' age, where children from older mothers score better in Battelle and Peabody in the short-term but these effects fade out in the medium-term.

[28]The analysis in terms of income has a caveat since income measured in 2017 could be endogenous to the earthquake, so the results should be taken cautiously.

[29]The continuous variables such as age and income per capita in 2017 are divided into two group based on the median of the variable.

Table 4: The Impact of the Earthquake on Cognitive and Non-Cognitive Development by gender

	(1)	(2)	(3)	(4)	(5)
		Short-term		Medium-term	
	Battelle	Peabody	CBCL1	Peabody	CBCL2
Panel A: Girl					
Affected*Earthquake	-0.0898	-0.0495	0.170	-0.0945	-0.0646
	(0.0913)	(0.0985)	(0.109)	(0.0817)	(0.110)
Observations	6,458	6,984	5,691	5,669	5,669
$Adj - R^2$	0.206	0.160	0.149	0.106	0.122
Panel B: Boy					
Affected*Earthquake	-0.128	-0.248**	0.0272	-0.257***	-0.225**
	(0.117)	(0.0982)	(0.116)	(0.0790)	(0.102)
Observations	6,583	7,194	5,944	7,120	5,879
$Adj - R^2$	0.230	0.135	0.146	0.201	0.110
p-value girl=boy	0.528	0.050	0.047	0.081	0.306

Notes: Each column represents a separate regression. The dependent variable is the test score measured in standard deviations. Peabody is the Peabody Test Score, Battelle is the Battelle Test Score and CBCL is the Child Behavior Check List measure. These scores are measured seven years later. Observations are at the individual/municipality level. Standard errors clustered at the municipal level appear in parentheses—number of clusters: 145. Short- and Medium-term mean measured two and seven years later. The estimates use sampling weights representative at the national level. Child/Mother/HH characteristics include gender, age at the moment of the test (in months), birth order, mothers' age and schooling, father present, number member of the household, prior mental health condition for mother, father and relative. P values come from a regression with a triple interaction between Earthquake, Affected, and Gender. All specifications contain cohort of birth and municipality fixed effects. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

5 Possible Mechanisms

In this section, I investigate three sets of possible mechanisms that could explain the negative impact on cognitive skills. First, I examine the effects on birth outcomes for children who were in utero during the earthquake. Second, I analyze child inputs relevant to the formation of human capital, such as health investments, breastfeeding practices, household income and general household instability. Third, I explore risky maternal behaviors, including smoking, drinking, and the use of recreational drugs during pregnancy and the first six months of the child's life and maternal stress. Although I may not be able to identify all mechanisms across all birth cohorts, I will test several relevant channels that may serve as pathways linking the earthquake to child development.

5.1 Birth Outcomes

In general, the literature has shown that negative shocks such as stress or poor health conditions during pregnancy have negative consequences on gestational age, birth weight, and height (Torche 2011, Aizer et al. 2016 B. Kim et al. 2017, Menclova and Stillman 2020). These middle outcomes have been shown that have a significant impact later in life such as in health, socioeconomic status, and mortality (Currie and Almond 2011, Torche 2018, Noghanibehambari 2022).

Table 5: The Impact of the Earthquake on Birth Outcomes

	(1) Weight at Birth	(2) LBW	(3) Height at Birth	(4) Birth weeks	(5) Preterm birth
In Utero*Earthquake	-0.0376	0.0104	-0.382*	-0.167	0.0115
	(0.0530)	(0.0207)	(0.221)	(0.192)	(0.0251)
Average	3.3937	0.0448	49.73	38.59	0.0946
Observations	14,479	14,479	14,378	15,317	15,317
$Adj - R^2$	0.0658	0.0715	0.0869	0.0732	0.0333

Notes: Each column represents a separate regression. Observations are at the individual/municipality level. Treated group is children in utero at the moment of the earthquake and the control group is the children born before. LBW: Low Birth Weight (<2.5kg). Preterm birth: less than 37 weeks of gestation. Standard errors clustered at the municipal level appear in parentheses—number of clusters: 145 in ELPI. The estimates use sampling weights representative at the national level. ELPI Child/Mother/HH characteristics include gender, age at the moment of the test (in months), birth order, mothers' age and schooling, father present, number member of the household, prior mental health condition for mother, father and relative. All specifications contain cohort of birth and municipality fixed effects. * p < 0.1, ** p < 0.05, *** p < 0.01.

In this section, I estimate the effect of in utero exposure to the 2010 Chilean earthquake on health indicators at birth using Equation 1[30]. Table 5 shows that only height at birth is statistically significant at conventional levels with an average reduction of 0.38 cms. To provide an assessment of the validity of this results, the raw trends of birth outcomes in Figure D.5 confirm that height at birth follows a parallel trend in the pre-shock years (2006–2009) and the early placebo window

[30]The measures at birth (weight, height and weeks of gestation) come from the Personal Child Health Record that it is a document for parents to keep a comprehensive record of their child's health, growth, and development

(2011-16) with some exception in 2012. The break is observed in the in utero cohort is unique to height, where earthquake exposure in the DiD in Table 5 leads to a statistically significant reduction. For other outcomes, including birth weight, and weeks of gestation the trends remain largely parallel across the majority of cohort-years. Pr(preterm birth) is a little bit noisier but the two series follow a similar trend. Low birth weight (LBW) is also a little noisier, the 2012 cohort deviates a lot from the trend, but other years move in parallel at some fair extent[31]. In general, the raw trends visually validate the estimates.

Overall, it is unlikely that these modest effects on birth outcomes alone account for the negative impact on cognitive and non-cognitive development in children who were in utero during the earthquake. This indicates that additional mechanisms are likely at play, warranting further exploration of other factors influencing child development post-earthquake.

5.2 Child Inputs

In this section, I explore factors relevant to human capital formation, including health investments such as the frequency of medical check-ups during pregnancy, vaccination rates, and breastfeeding duration, household income, assets shocks and overall household environment all of which may have been impacted by the earthquake.

Health Inputs. The occurrence of an earthquake can severely damage public infrastructure, such as hospitals, which in turn affects access to both pre- and post-natal care. At the same time, breastfeeding is recognized as a crucial factor in the development of health and human capital. The disruption caused by an earthquake can interrupt breastfeeding for several reasons. First, to mitigate the negative effects of the disaster, mothers may need to enter the labor market to generate income. Second, stress related to the earthquake can impact breastfeeding by increasing cortisol levels in breast milk, which can influence the maternal organization of bio-behavioral processes in offspring (Dettmer et al. 2018)[32]. Additionally, breastfeeding may stop due to damage to infrastructure, which can compromise maternal health through increased risks of infectious diseases, contaminated water, or lack of access to clean water.

Table 6 presents the results. In Column 1, the effect on the number of vaccinations is not statistically significant. Columns 3 and 4 display the impact on the probability of attending prenatal check-ups and the total number of check-ups, both of which are small and not statistically significant. Lastly, Column 2 indicates that the effect of the earthquake on the duration of breastfeeding is

[31] One reason to the lack of significant results is that the existing literature identifies very small effects on birth outcomes. To accomplish this, these studies frequently utilizes universal population registries (e.g., all births in a country). These datasets provide the massive sample sizes necessary to achieve high precision, indicated by low standard deviations, required to identify effects that are often small in magnitude. For this reason, I carry out an additional analysis using the Vital Records data of all births in Chile between 2008 and 2015. In the appendix, the Table C.39 shows significant and negative effects on height at birth, though economically modest, an average reduction of 0.125 cms in the earthquake municipalities. In general, the estimates from Vital Records do not deviate from the point estimates in ELPI in sign and significance.

[32] This phenomenon is known as the lactational programming hypothesis, which posits that the biological constituents of breast milk influence metabolic and neurobiological development, subsequently affecting offspring phenotype and behavior (Hinde 2013).

small, positive, and non-statistically significant. The raw trends in Figure D.6 validate the estimates for number of vaccines, breastfeeding and the two measures of check-ups [33].

Table 6: The Impact of the Earthquake on Health Care Investment

	(1)	(2)	(3)	(4)
	Post-Natal Care		Pre-Natal Care	
	# Vaccines	Breastfeeding	Pregnancy Check-up (=1)	# Pregnancy Check-ups
Treated*Earthquake	0.007	0.826	0.006	-0.002
	(0.011)	(0.548)	(0.020)	(0.610)
Average	5.981	14.33	0.984	10.63
Observations	15,679	14,126	15,686	15,676
$Adj - R^2$	0.010	0.117	0.073	0.063
Treated	In Utero-4yo	In Utero-4yo	In Utero	In Utero
Control	Unborn	Unborn	Alive-Unborn	Alive-Unborn

Notes: Each column represents a separate regression. Breastfeeding is measured in months. Observations are at the individual/municipality level. Standard errors clustered at the municipal level appear in parentheses—number of clusters: 145. The estimates use sampling weights representative at the national level. Child/Mother/HH characteristics include gender, age at the moment of the test (in months), birth order, mothers' age and schooling, father present, number member of the household, prior mental health condition for mother, father and relative. All specifications contain cohort of birth and municipality fixed effects * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Household Income. Next, I move on to analyze the effect of the shock on household per capita income. Household income has been shown that has a positive effect on early child health and cognitive development (Duncan et al. 2011, Dahl and Lochner 2012, Hoynes et al. 2015). To estimate the effect of the earthquake on household income two years after the disaster, I use the ELPI sample of children who were alive at the moment of the earthquake[34] and for whom pre-earthquake information is available from the first wave of data collection. I employ a DiD approach exploiting geographic and household time variation rather than geographic and birth cohort variation. This allows to compare household income before and after the earthquake between the most and least affected municipalities two years later. A caveat to this analysis is that the post-earthquake income is measured two years after the shock, it may mean that any immediate effects could have dissipated by that time. Additionally, using the same source of data, I look at the change in mother labor status in the households of the children who experienced the earthquake two years after the shock.

[33]The placebo years are cohort 2011-16. I measure health outcomes up to age three for the relevant cohorts, as this period encompasses the most critical developmental milestones and standardized health interventions in Chile. This time frame aligns with the national 'Well-Child' checkup schedule (known as Control de Niño Sano), which involves frequent clinical assessments at birth, 7–10 days, and monthly or bi-monthly intervals through the first year, followed by visits at 18, 24, 30, and 36 months. These checkups are strictly integrated with the National Immunization Program (PNI), covering the BCG, Hexavalent, Pneumococcal, and MMR vaccines, and provide standardized metrics for breastfeeding. In Chile, while exclusive breastfeeding has a 59% prevalence at six months, the mean duration is 11.7 months, with a notable transition to solid foods occurring between 12 and 18 months. Focusing on the first 36 months thus ensures a robust and comparable dataset of physical growth, immunization status, and nutritional transitions across the sample.

[34]These children were between 7 and 48 months old or cohorts from in utero to 4 years-old.

Table 7: The Impact of the Earthquake on Household Income Per-Capita and Mothers' Labor Force Participation

	(1)	(2)	(3)	(4)
	Two Years Later		Two Months Later	
	Income pc	Mothers' LFP	Income pc	Mothers' LFP
Earthquake*Post	-6,524.70	0.0027	-12,832.82**	-0.0538**
	(6,763.6)	(0.0141)	(4,943.1)	(0.0229)
Average	105,209.70	0.44	127,347.80	0.391
Observations	23,384	23,384	8,562	8,562
$Adj - R^2$	0.0621	0.0329	0.248	0.0824

Notes: Each column represents a separate regression. Outcomes measured two years later comes from the ELPI survey and income measured two months after comes from the CASEN Earthquake Panel. LFP: Labor force participation. Observations are at the household/municipality level. The estimates use sampling weights representative at the national level. These estimates are a difference-in-difference comparing outcomes from household present in the baseline ELPI survey pre- and post-earthquake and comparing pre (Post=0) and post (Post=1) income of household with at least one child younger than four years old in the the CASEN Earthquake Panel. Income pc is Income per capita. All specifications contain municipality and year effects. Standard errors clustered at the municipal level appear in parentheses—number of clusters: 145. * p < 0.1, ** p < 0.05, *** p < 0.01.

To analyze the effect on the short-term income or the immediate impact, I use an additional source of information, the CASEN Earthquake Panel 2009-10. This is a nationally representative panel of households that measures demographic, income, and earthquake information before and three months after the shock. I focus on the household that had at least one child under four years old[35] to evaluate the change in income caused by the earthquake.

Column 1 of Table 7 indicates that two years following the earthquake, there is a non-significant reduction of 6% on household income per capita. Column 2 reveals that the employment status of mothers did not change because of the earthquake. Column 3 details the immediate financial impact of the shock, showing a 10% reduction in per capita household income in affected municipalities. This income reduction may be largely driven by a sharp 5 percentage points drop in maternal labor force participation, a 12% decline relative to the baseline.

Structural Asset Shocks, Resource Reallocation, and Child Daycare Enrollment. While the mothers' labor market exit directly explains the immediate income loss, it also signals a profound shift in household dynamics that may be more damaging to a child than the financial deficit itself. It is crucial to distinguish between the temporary *income flu*, a 10% drop that recovered within two years, and the structural asset shock. Beyond the flow of monthly income, the earthquake represented a massive shock to household wealth and stability.

The evidence from the CASEN Earthquake panel (Table 8, Columns 1 to 2) indicates a robust association between seismic intensity (PGA) and minor and major damage to the house. While these estimates are cross-sectional, the identification strategy exploits the unpredictable and exogenous

[35]The demographics and income characteristics pre-earthquake between the ELPI and CASEN Earthquake Panel are shown in Table C.9 and are similar for this group of families with at least one child under four years old.

physical variation of seismic waves. To mitigate concerns regarding endogenous sorting or baseline vulnerability, I demonstrate that PGA is uncorrelated with 2009 household characteristics, including income, mothers' education and age, father presence and pre-existing house quality, supporting the assumption that the shock's intensity was effectively random (see Appendix Table C.10).

Table 8: Household Damage and Recovery Costs: Suggestive Evidence

	(1) *Major Damage* (=1)	(2) *Minor Damage* (=1)	(3) *Repair Cost* (CLP)	(4) *Own Resources* (=1)
Panel A: Binary Impact				
Earthquake Exposure	0.087***	0.326***	98,577***	0.102***
	(0.011)	(0.030)	(18,326)	(0.019)
Observations	4,275	4,275	4,275	4,275
Adj. R^2	0.038	0.105	0.014	0.029
Panel B: Seismic Intensity				
PGA Intensity	0.378***	1.238***	484,710***	0.438***
	(0.052)	(0.087)	(77,470)	(0.063)
Observations	4,118	4,118	4,118	4,118
Adj. R^2	0.061	0.165	0.029	0.053

Notes: Each column reports coefficients from OLS regressions. *Earthquake* is an indicator for affected municipalities; *PGA* denotes Peak Ground Acceleration (m/s^2). Columns 1 and 2 are binary indicators for damage severity. Column 3 reports the self-reported cost of household (HH) repairs in Chilean Pesos (CLP). Column 4 is a dummy indicating repairs were financed using the household's own funds. All models include sampling weights and control for household size, maternal age, education, paternal presence, number of children, and 2009 baseline per capita income, house quality, and assets. Robust standard errors (in parentheses) are clustered at the municipality level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

This structural asset shock likely initiated a period of residential and domestic instability that disrupted the child's environment during their most vulnerable developmental stage. Column 3 shows that the average cost of repair in impacted households was approximately CLP 98,577, an amount nearly equal to the average monthly per capita income of these households. Finally, Column 4 indicates that households in high intensity municipalities were 10 percentage points more likely to exhaust their own resources for repairs, suggesting a significant diversion of liquid assets away from child-centered investments to finance physical recovery.

This structural asset loss likely functioned as a persistent stressor, compounding the psychological strain analyzed in the next Section 5.3. Within this framework, physical destruction serves as a tangible trigger for the disrupted domestic environment and the elevated maternal stress documented across both datasets (Tables 10 and 11 in the next Section 5.3). Beyond the psychological burden, Columns 3 and 5 of Table 10 show that the shock also had direct physiological consequences: mothers in high intensity municipalities were 2.3 percentage points more likely to report earthquake-related health complications, even after accounting for baseline socio-economic status and household

characteristics (Columns 3 to 4 in Table 10). These findings suggest that the disaster's impact was not merely a transitory event, but a multifaceted disruption to the maternal internal environment that persisted long after the tremor.

In this context, the mothers' exit from the workforce and the deterioration of the home act as complementary markers for the toxic stress[36] that characterizes the post-disaster environment. This shock operated as a stress budget trap: the household likely was forced to prioritize immediate physical recovery over the unobserved quality of the domestic environment. This framework allows for a unified explanation of the paper's main results: the earthquake triggered a biological and resource scar for the in utero to 6 months old cohorts and a structural resource scar for the children from 0 to 4 years-old, both of which manifest as persistent human capital deficits in middle childhood.

A critical question is whether these disruptions manifested as a withdrawal from formal child care or schooling. However, estimates in Table C.11 using the CASEN Earthquake Panel to short-term (three months later) and ELPI to medium-term impact (two years later) show that enrollment for both young children (0-4 years) and older cohorts remained remarkably stable despite the shock. This suggests that the developmental setbacks were not driven by a lack of access to educational institutions, but rather by a reallocation of resources within the household.

While families maintained the quantity of investment through stable school attendance or health care (vaccines or medical check-ups), the sudden need to repair physical assets, costing approximately one month's per capita income likely forced a reduction in the early caregiving environments and attachment quality including nutritional inputs. This trade-off in nutrition is well-documented in the literature; when households face severe asset shocks but keep children enrolled in school, they often buffer the educational cost by reducing expenditures on high-quality proteins and micro-nutrients. For instance, studies in similar Latin American contexts, such as Rosales-Rueda 2018 in Ecuador and Aguilar and Vicarelli 2022 in Mexico, demonstrate that income drops of similar magnitudes triggered significant reductions in food and nutritional spending even when schooling remained constant. Consequently, the multi-dimensional disruption of the early caregiving environment, where household resources were diverted from optimal nutrition toward immediate recovery, provides a clear precedent for the medium-term cognitive and non-cognitive deficits.

5.3 Mothers' Risky Behaviors and Stress

A large body of medical research documents a robust association between exposure to traumatic events, such as natural disasters, and increases in cigarette, alcohol, and recreational drug consumption as coping mechanisms among affected populations (DiMaggio et al. 2009, Do and Iyer 2012, Friedman 2020). A separate but related literature shows that these risky behaviors adversely influence a wide range of child outcomes, including health, cognitive and non-cognitive development, and longer-term indicators such as educational attainment and labor market performance (Chatterji

[36]According to Franke 2014 toxic stress refers to the excessive, long-lasting activation of the stress response system in the body, occurring when a child faces severe, frequent adversity (e.g., abuse, neglect, poverty) without adequate support from caring adults. This, unlike positive stress, can alter brain development and increase risks for chronic physical and mental health issues.

and Markowitz 2001, Nilsson 2017, Mangiavacchi and Piccoli 2018, Kreif et al. 2022). This section examines whether maternal risky behaviors represent one empirically observable pathway through which exposure to the 2010 Chilean earthquake may have affected early child development. I analyze these behaviors as a potential mechanism rather than an alternative explanation, viewing them as part of a broader bundle of negative shocks that includes structural asset loss, household instability, and parental stress.

In particular, parental smoking has been consistently linked to detrimental effects on children's health (Mellor 2011, Settele and Van Ewijk 2018, McGeary et al. 2020). The medical and economic literature identifies birth weight as a primary channel for these effects; Lushniak et al. 2014 estimate that maternal smoking reduces an infant's birth weight by between 200 and 300 grams, while W. N. Evans and Ringel 1999 demonstrate that infants born to mothers who quit smoking due to policy interventions experienced an average birth weight gain of approximately 400 grams. Notably, Simon 2016 shows that exposure to smoking in utero and during early infancy entails substantial medium-term health costs, such as a 10% increase in school sick days. In the domain of cognitive and non-cognitive development, Srivastava and Trinh 2021 provide causal evidence that parental smoking reduces children's test scores and worsens behavioral outcomes.

In Chile, the smoking culture during the 2000 to 2015 period was marked by one of the highest smoking prevalence in Latin America. A defining characteristic was the narrow gender gap in smoking rates, a deviation from the regional norm where male smoking typically far exceeds female smoking. This pattern highlights a significant cultural susceptibility and acceptance of smoking among women, who consistently registered prevalence rates peaking above 35% in the early 2010s (Nazzal and Harris 2017)[37]. This high cultural acceptance translates directly to smoking during critical reproductive stages. Studies conducted during this period estimated the prevalence of active smoking during pregnancy in vulnerable, low-socioeconomic status populations to be extremely high, around 28% (Mallol et al. 2007)[38].

To estimate the impact of the earthquake on risky mother behaviors, I focus on mothers who were pregnant or had a child from zero to six months of life at the moment of the earthquake using information from the ELPI survey using Equation 1. I report estimates using a binary outcome variable that equals one if the mother smokes or drinks or uses recreational drugs during this stage, and zero otherwise. Additionally, I construct a composite variable that equals one if she engages in at least one of these behaviors.[39].

[37]In response to this crisis, the Chilean government introduced two major pieces of legislation during this time: the 2006 Tobacco Law and the far more comprehensive 2013 Tobacco Control Law. The 2013 law, which implemented a nearly complete ban on smoking in all indoor public spaces (including workplaces and restaurants), began the crucial process of de-normalizing smoking culture and driving down the historically high overall prevalence rates. Since no municipal smoking bans reforms coincide with Chile's 2010 earthquake, local policy shifts do not confound the results. The study instead exploits geographic earthquake intensity, which remains uncorrelated with national legislative timelines.

[38]This study is based on one specific hospital in Santiago, Chile in middle 2000 where low income mothers usually attend. In my case, smoking during pregnancy is around 11% and there is no difference between education level or household income.

[39]Self-reported measures of risky behaviors and early-life practices may suffer from imperfect recall, especially when mothers report events that occurred several years earlier. In a DiD setting, such recall error only biases the estimates

Table 9: The Impact of the Earthquake on Mothers' Risky Behaviors of children in utero and up to 6 months old

	Risky Behaviors				
	(1)	(2)	(3)	(4)	(5)
	Any	Smoke		Alcohol	Drug
		Extensive	Unconditional Intensity		
Treated*Earthquake	0.0798**	0.0458*	-0.1915	0.0564	0.0302
	(0.0363)	(0.0256)	(0.4458)	(0.0369)	(0.0217)
Average	0.1903	0.1107	4.883	0.1082	0.0155
Observations	15,412	15,412	15,412	15,412	15,412
$Adj - R^2$	0.0645	0.0624	0.2061	0.0626	0.0696
Treated	In Utero 0 to 6 mo	In Utero 0 to 6 mo	In Utero 0 to 6 mo	In Utero 0 to 6 mo	In Utero 0 to 6 mo
Control	$>$ 6 mo Unborn	$>$ 6 mo Unborn	$>$ 6 mo Unborn	$>$ 6 mo Unborn	$>$ 6 mo Unborn

Note: Each column represents a separate OLS regression, except for Column (3). Column (1) "Any" is a binary indicator that takes the value of 1 if the mother reported smoking or drinking or using recreational drugs during pregnancy or the child's first six months of life. Column (2) "Extensive" is a linear probability model for the binary decision to smoke ($Smoke > 0$). Column (3) "Unconditional Intensity" measures the number of cigarettes consumed per month; due to the presence of zero-consumption households and the skewed distribution of tobacco use, this column is estimated using a Poisson Pseudo-Maximum Likelihood (PPML) model, where coefficients are interpreted as semi-elasticities (percentage changes). Columns (4) and (5) are linear probability models for alcohol and drug consumption, respectively. The Treated group includes children who were in utero or up to six months old at the time of the 2010 earthquake; the Control group consists of children who were at least seven months old. All specifications include municipality and birth-cohort fixed effects, and control for maternal age, schooling, fathers' presence, household size, and pre-existing mental health conditions. Standard errors are clustered at the municipal level (145 clusters). Sampling weights are applied to ensure national representativeness. * p $<$ 0.1, ** p $<$ 0.05, *** p $<$ 0.01.

As shown in Table 9, the estimates indicate a significant increase in maternal risky behaviors attributable to earthquake exposure. Column 1 shows that the probability of a mother engaging in any risky behavior increased by 8 percentage points (pp), representing a 42% relative change. When examining specific behaviors, Column 2 reveals that mothers were 4.6 pp more likely to smoke, a substantial 41% relative change[40]. This result is consistent with the medical and economic evidence on behavioral responses to stress including consumption of cigarettes and other substances as coping responses (DiMaggio et al. 2009, Do and Iyer 2012, Friedman 2020).

To put the 41% increase in smoking in perspective, W. N. Evans and Ringel 1999 find that a 10% tax hike, a major policy intervention, typically reduces smoking participation by only about 5% in pregnant women. The earthquake's impact is nearly eight times that magnitude. While this disparity is large compared to price-based elasticities, it is consistent with the difference between a marginal pecuniary change and an acute traumatic shock.

Column 3 of Table 9 examines the smoking unconditional intensity, measured by the number of cigarettes smoked per month among mothers[41]. The estimated coefficient in the full sample is negative (-0.19 or -19%), imprecise, and statistically indistinguishable from zero. I interpret this null result as a reflection of substantial variance in cigarette consumption driven by a small number of extreme outliers at the top of the distribution. To investigate this further, I provide a robustness check in Appendix Table C.41 (Columns 3 and 4) that excludes the top 1% of the smoking intensity distribution (those consuming ≥ 70 cigarettes per month). In this outlier-corrected specification, the estimated effect on unconditional intensity becomes positive and statistically significant (0.825), indicating that the earthquake increased total tobacco exposure within the light-to-moderate consumption range. Crucially, as shown in Columns 1 and 2 of Appendix Table C.41, the effect conditional on smoking ($Smoke > 0$) remains statistically insignificant (-0.36).

This suggests that the behavioral response to the earthquake operated almost exclusively through the extensive margin (smoking initiation or relapse) rather than an increase in consumption among pre-existing heavy smokers. This interpretation is consistent with prior evidence showing that acute traumatic shocks primarily affect smoking participation (Ayyagari and Sindelar 2010, Iakunchykova et al. 2015). Furthermore, results shown in Columns 1 and 2 in the Appendix Table C.41 reflect the potential treatment-selection of the $Smoke > 0$ sample: the entry of new, lighter smokers into the treated group likely masks any potential intensive-margin shifts by pulling the conditional average

if the accuracy of recall changes differentially across treated and control municipalities after the earthquake. If recall accuracy is similarly imperfect across groups, the resulting non-systematic error behaves classically and attenuates coefficients toward zero, making effects harder to detect. If the earthquake affected recall accuracy unevenly across groups, bias could in principle go in either direction. In this study, I do not suspect of suggestive evidence of differential misreporting correlated to treatment status. For instance, the Figure D.4 in the appendix shows the raw trends for risky behavior index, smoking and drinking; they do not show large deviation for earthquake and non-earthquake municipalities across cohorts that are used as placebo (2006-09 and 2011-16), specially for risky behavior index and smoking, and at some extent, drinking. Therefore, this concern is acknowledged as a general limitation of self-reported data rather than a threat.

[40] Figure D.4 validates the parallel trend assumption of risky behavior index and smoking.

[41] The unconditional intensity combines the extensive and intensive margins. A pure intensive margin estimate is not feasible here because the treatment may influence the composition of the smoking population (selection bias) and individual-level baseline smoking status is unavailable in the repeated cross-sections.

downward.

Importantly, the clinical relevance of these extensive-margin shifts does not depend on changes in smoking intensity. The medical consensus establishes that no safe level of tobacco exposure exists in utero, as even limited exposure induces placental vasoconstriction and fetal hypoxia (Lushniak et al. 2014). Consequently, the marginal harm of smoking during pregnancy is highest at low levels of consumption (DeCicca et al. 2022, Lakdawala and Simon 2017).

Additionally, maternal stress also plays a relevant role, estimates from the 2010 CASEN Earthquake panel indicate that mothers in the most impacted municipalities were 9 percentage points more likely to report clinical indicators related to stress including insomnia and panic attacks (Table 10). While these estimates are cross-sectional, they leverage the same exogenous variation in seismic intensity used to identify household damage, where I previously demonstrated that intensity is uncorrelated with baseline characteristics (Appendix Table C.10).

To further isolate the role of stress in the caregiving environment, I estimate a DiD model using the 2012 ELPI wave, comparing mothers whose children were alive during the tremor (2006–2010 cohorts) to those who became mothers in 2011. This approach exploits the timing of motherhood relative to the shock. The results in Table 11 suggest that motherhood significantly amplified the earthquake's psychological toll, increasing reported clinical stress by 2 percentage points. This provides suggestive evidence that the caregiving burden served as a primary channel for the persistent stress response observed in the data.[42]

While the observed behavioral response, mediated by smoking and stress, to the shock appears relatively short-lived, its temporal alignment with the peak of fetal and neonatal neuroplasticity provides a direct biological pathway for long-term damage in the youngest cohort. For children exposed in utero or during the first six months of life, even a temporary disruption can trigger a process of biological scarring during this sensitive developmental window. Following the *Fetal Origins Hypothesis* (Almond and Currie 2011), environmental shocks during critical gestation windows program biological systems in ways that permanently alter neuro-developmental trajectories (Shonkoff et al. 2012; Cunha and Heckman 2007).

Furthermore, this increase in smoking participation occurs despite a simultaneous reduction in household income and mothers' labor-force participation documented in Table 7[43]. Crucially, the persistent cognitive and non-cognitive deficits observed seven years later do not appear to stem

[42]This analysis relies on retrospective self-reports from the 2012 ELPI wave including insomnia, panic attacks, and traumatic memories; details on these indices are provided in the notes for Table 11. While the single post-treatment cohort precludes a formal parallel trends test, the results remain consistent across various clinical threshold definitions. Due to the timing of the survey, this exercise cannot be expanded to the 2017 refreshment cohort.

[43]Increases in maternal smoking are consistent with the medical and economic evidence on behavioral responses to stress. Research on traumatic events (including earthquakes) documents elevated PTSD risk and higher consumption of cigarettes and other substances as coping responses (DiMaggio et al. 2009, Do and Iyer 2012, Friedman 2020). While these studies on natural disasters often leave the specific impact of income loss implicit, the observed persistence or increase in smoking, even amidst tightened budget constraints, suggests that the psychological need for stress-coping frequently overrides the economic pressure to reduce consumption. Evidence from job-loss settings, where income falls sharply and stress rises, further shows that smoking often increases despite reduced budgets (Everding and Marcus 2020, Marcus 2014, Falba et al. 2005). This pattern reflects the low income elasticity of tobacco demand and the tendency to increase smoking under stress, even when budgets become tighter.

Table 10: Maternal Stress and Health Complications: Descriptive Evidence (CASEN Earthquake Panel)

	(1)	(2)	(3)	(4)
	Stress		*Health Issues*	
	Binary	Intensity	Binary	Intensity
Earthquake Exposure	0.092***		0.023***	
	(0.023)		(0.005)	
PGA Intensity		0.414***		0.097***
		(0.067)		(0.026)
Observations	2,600	2,600	4,118	4,118
$Adj - R^2$	0.018	0.036	0.020	0.025

Notes: Each column represents a separate OLS regression using the CASEN Earthquake Panel. *Earthquake* is a binary indicator for affected municipalities; *PGA* denotes Peak Ground Acceleration (m/s^2). *Clinical Stress* is a binary indicator based on the Davidson Trauma Scale. *Health Issue* is a self-reported health problem directly caused by the earthquake. All models include sampling weights and controls for household size, maternal demographics (age, education), paternal presence, number of children, and 2009 baseline per capita income and house quality. Robust standard errors (in parentheses) clustered at the municipality level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 11: Impact of Earthquake Exposure on Reported Maternal Stress (DiD Estimates)

	(1)	(2)	(3)	(4)	(5)	(6)
	Severity Threshold		*Robust Syndromic*		*Clinical Core*	
	Binary	Intensity	Binary	Intensity	Binary	Intensity
Affected × Earthquake	0.020**		0.019***		0.021***	
	(0.009)		(0.007)		(0.007)	
Affected × PGA		0.090***		0.081***		0.090***
		(0.033)		(0.031)		(0.031)
Observations	14,106	14,106	14,106	14,106	14,411	14,106
$Adj - R^2$	0.067	0.067	0.049	0.050	0.050	0.050

Notes: Estimates from a Difference-in-Differences model using the 2012 ELPI wave, comparing mothers of the 2006–2010 cohorts (treated) to the 2011 cohort (control). The following are the possible symptoms: insomnia, stress or anxiety, emotional instability, panic attacks, and persistent traumatic memories. *Severity Threshold* is an indicator for a symptom sum ≥ 3 (PC-PTSD-5); *Robust Syndromic* requires a sum ≥ 3 plus traumatic memories; *Clinical Core* captures traumatic memories plus panic attacks/insomnia. *PGA* denotes Peak Ground Acceleration (m/s^2). All models include sampling weights and controls for household size, maternal age, education, paternal presence, number of children, and 2009 per capita income. Robust standard errors (in parentheses) clustered at the municipality level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

from a breakdown in basic health service provision or a lack of fundamental maternal health-seeking behavior. As demonstrated in Section 5.2, the earthquake had no significant impact on standard health inputs, such as breastfeeding duration, vaccination rates, or the frequency of pre- and post-natal medical check-ups (Table 6) as well as daycare enrollment (Table C.11). This suggests that the institutional quantity of care remained stable, and that the primary channel of impact operates through more nuanced, within-household disruptions.

While household economic conditions recover within two years, the persistence of developmental deficits for children from in utero and up to six months-old cohorts suggests an impact beyond temporary deprivation. This pattern, characterized by biological and psychological scarring, is mirrored in the low income elasticity of tobacco demand, where the need for stress-coping overrides budget constraints. Ultimately, the observed increase in smoking serves as a visible, clinically relevant biomarker for a broader bundle of household trauma, combined with the immediate shocks of income loss, reported stress and the urgent burden of repairing physical assets discussed in section 5.2, that underpins these enduring developmental gaps.

Finally, Columns 4 and 5 of Table 9 indicate that mothers in the most affected municipalities exhibit higher probabilities of alcohol consumption and recreational drug use, although these estimates are not statistically significant[44]. Visual inspection of Figure D.4 suggests that the parallel trends assumption does not hold for these outcomes, limiting the interpretation of these estimates.

6 Conclusion

Natural disasters impose a profound cost on society, particularly during the sensitive stages of early development. While the immediate physical impacts of such shocks are well-documented, this paper provides rare evidence on how unpredictable seismic events alter the medium-term developmental trajectories of children.

Seven years after the 2010 Chilean earthquake, I find that children exposed in utero and up to age four experienced a significant decline of 0.17 SD in cognitive skills and 0.13 SD in non-cognitive indicators. This reduction is economically substantial, representing an equivalent loss of approximately 0.8 years of schooling. These findings are robust to a wide battery of tests, including alternative intensity measures, bounding techniques for differential attrition and event studies estimates that support the parallel trend assumption.

The persistence of these deficits points to a multi-dimensional disruption of the early caregiving environment. I find that the damage is not driven by a collapse in formal health investments, but by a combination of toxic stress and asset reallocation that affected children at different developmental stages. For the in utero and up to 6 months old, maternal smoking served as a visible biomarker for a broader internal environment shock that triggered permanent biological scarring as well as

[44]Additionally, I carry out an analysis using as a control group only children who are at least seven months old at the moment of the earthquake to alleviate any concern that the control group using children conceived after the earthquake is contaminated. Table C.40 shows the estimates in the same magnitude but less precise since the sample reduces about 37%.

resource shock. Simultaneously, for the older toddlers, the sudden need to finance home repairs, costing a full month of per capita income, forced a structural trade-off. Households successfully preserved the quantity of care (vaccinations and school enrollment) but were forced to sacrifice the quality of the domestic buffer and unobserved home-care inputs to finance physical recovery. This stress-budget trap effectively shifted children across the entire early-childhood spectrum onto a lower developmental trajectory that remains detectable seven years later.

From a public policy perspective, these findings are particularly timely and relevant. As noted by Almond et al. 2018, measuring medium-term outcomes resulting from early childhood exposure to adverse shocks is crucial for understanding developmental trajectories and informing policy responses. Such evidence can support governments in designing and implementing timely interventions to mitigate the long-term consequences of increasingly frequent natural disasters. Importantly, this study highlights that the effects of early-life shocks extend beyond cognitive development to also include non-cognitive outcomes, a dimension that has been largely under-explored in the context of earthquakes in developing countries. Furthermore, the results underscore the need for institutional attention to the well-being of parents in the aftermath of natural disasters. Parental stress can lead to increased engagement in risky behaviors such as smoking, alcohol consumption, and drug use, which are commonly used as coping mechanisms in high-stress environments. These behaviors, in turn, may have adverse consequences for children's development, particularly during critical early life stages.

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A Appendix: Contextual Background of the 2010 Maule Earthquake

A.1 Geophysical Context and Immediate Impact

Chile is located along the boundary between the Nazca and South American tectonic plates, one of the most seismically active subduction zones in the world. The continuous convergence of these plates generates the accumulation of tectonic stress that is periodically released through large-magnitude earthquakes, often accompanied by tsunamis. The intensity of shaking experienced by the population is also modulated by local geological conditions, specifically the type of ground and soil in each municipality. Despite the nation's strict seismic construction codes, the M_w 8.8 megathrust earthquake of February 27, 2010, overwhelmed regional resilience. It remains the second-strongest in Chilean history and was, at the time, ranked as the world's sixth-largest recorded seismic event. The resulting rupture, spanning roughly 540 by 200 km, displaced the seafloor and generated a catastrophic tsunami that impacted over 500 km of coastline, with run-up heights reaching up to 29 meters in coastal areas like Constitución (Larrañaga and Herrera 2010).The scale of the physical destruction was immense. The combined event directly affected two million residents, resulting in a confirmed death toll of 547 casualties (Muñoz 2010, Contreras and Winckler 2013). Infrastructure damage was widespread: approximately 550,000 housing units were severely affected, including 370,000 homes deemed destroyed or uninhabitable. Critical infrastructure suffered similarly, with over 4,200 educational facilities damaged, close to one-third of facilities in the affected regions, while over 70% of the national hospital network experienced some level of impact, including the loss of more than 4,000 hospital beds (Reconstrucción 2015).

A.2 Aftershocks and Exposure Duration

The earthquake was not a singular event but a persistent shock. The affected regions experienced thousands of aftershocks following the main event in the subsequent month (March 2010) , including several exceeding M_w 6.0 (López Tagle and Santana Nazarit 2011). These episodes prolonged displacement, delayed infrastructure recovery, and sustained high levels of stress and uncertainty among households. As a result, the earthquake constituted both an acute trauma and a chronic environmental stressor that extended well beyond the initial event in February 2010.

A.3 Relief Efforts and Reconstruction Timeline

The government response, formalized as the *Reconstruction Plan 2010*, targeted displaced households, public service users, and local firms. Housing policies were designed to provide permanent solutions; by 2014, more than 240,000 housing subsidies were allocated nationwide (Reconstrucción 2015, Lunecke 2016). Public investment also targeted the demolition of high-risk buildings and the restoration of transport networks and water systems. Economic support measures focused on small and medium-sized firms to stabilize employment, supporting 42,000 enterprises through asset repair subsidies (Reconstrucción 2015).By the mid-2010s, a significant portion of reconstruction goals remained incomplete, particularly in housing. Reports indicated that only about 24% of damaged homes had been successfully rebuilt by 2011 (Comerio 2013). This led to the implementation of a formal *Reconstruction Closure Plan* in 2014, which placed greater emphasis on the social dimensions of recovery, explicitly addressing the psychosocial consequences of the disaster to mitigate trauma and social dislocation among affected populations.

A.4 Institutional Recovery and the 2011 Cohort

A critical element of the identification strategy is the rapid recovery of essential human capital infrastructure relative to the housing sector. Assessments by CEPAL 2010 indicate that 92.7% of health facilities suffered minor or no damage, with the majority becoming fully operational by June 2010 (Government of Chile 2010). Structural damage was highly concentrated, with 44% of damaged establishments located in the Maule region.[45]Furthermore, according to the Ministry of Education and the National Kindergartens Board (JUNJI), schools and public daycare centers resumed operations by April and May 2010, respectively. Consequently, the 2011 cohort benefited from a functional healthcare and educational environment. This distinguishes them from the 2006–2010 cohorts, who faced acute developmental shocks during their critical windows of development, characterized by severe maternal stress, home destruction, and the temporary breakdown of local support systems.

A.5 Global Significance and Long-term Human Capital Costs

Beyond the immediate humanitarian impact, large earthquakes warrant attention due to their massive indirect economic costs. Globally, earthquakes of magnitude 7.0-7.9 occur 10-20 times per year, while events of magnitude 8.0 or higher occur once annually (Earthquake Hazards Program 2023). Since 1900, at least twelve earthquakes have resulted in more than 50,000 fatalities each, and geophysical disasters have generated an estimated USD 1.6 trillion in losses between 1970 and 2023 (Disaster Risk Reduction 2025).In Chile, the 2010 disaster resulted in losses of approximately USD 30 billion, or 18% of GDP (Magud 2010). While direct physical damages account for 50-70% of observed losses, a growing literature shows that indirect effects, such as those identifying the impact of stressors on fetal and early childhood development, reveal costs that unfold over decades. Empirical evidence links early exposure to severe stressors to lower birth weights and persistent reductions in cognitive achievement and adult earnings. The 2010 Maule earthquake provides a uniquely relevant setting to study these human capital consequences.

B Appendix: Event Cohort Study Analysis

B.1 2015 Peabody Coefficient

In the main estimates, I define the affected cohort 2006-2010 and the placebo 2011-15. I choose as a baseline the 2011 cohort for the event cohort study. It is important to note that an issue arises from splitting the sample into cohort, the event cohort study coefficients becomes noisier. Also, these large confidence intervals do not allow to find heterogeneous effect by age since all the interval intersect themselves.

Figures 2 and 3 show the event cohort studies for Peabody test in the short and medium-term. The coefficients for the 2012–2014 cohorts are statistically indistinguishable from zero, therefore the parallel trend assumption holds for these cohorts. However, the 2015 cohort shows a negative, large and significant coefficient, the youngest and furthest post-treatment cohort.

To investigate this, first, I look at the Peabody raw trends. In figures D.2 and D.3, the gap between Earthquake and Non-Earthquake municipalities remains constant at some reasonable extent for the placebo cohorts. From 2011 to 2014, the two groups follow an identical upward slope in raw cognitive scores as children age, something that is reassured in the event studies. This consistency in the gap during these four placebo years provides strong empirical support for the parallel trends

[45]Regressions excluding the Maule Region yield results consistent with the main specification.

assumption required for the DiD identification. However, in 2015 the trends diverge and cross each other, even changing the order of the series.

Next, I provide extensive evidence about the reason that the raw series suddenly change their pattern that ended up in a significant coefficient estimate. I argue that this coefficient is not a failure of the identification strategy, but rather it might be a measurement artifact resulting from psychometric irregularities in the testing of 2 year old (the 2015 cohort). We justify the treatment of the 2015 cohort as a boundary case based on the following criteria:

- **Psychometric Failure and 'Floor Effects' in the 2015 Cohort**. The 2015 cohort represents the lower age bound of the sample (age 2.5 at test). The raw data in Tables B.1 and B.2 reveals three irregularities suggesting the that the Peabody lacks the resolution to capture meaningful variation at this age:
  - Mechanical Mean Inflation: For age-standardized scores, the mean should remain stable across cohorts. At first glance, the 2015 cohort exhibits a mean (105.59) that appears comparable to the 2014 cohort (104.58). However, this similarity is actually a mathematical byproduct of the floor effect rather than a reflection of true cognitive ability. By effectively censoring the lower tail of the distribution, where the minimum score for the 2015 cohort is 80 compared to 66 for the 2014 cohort, the mean is mechanically forced upward. This lack of downward variation, which deviates from the score stability seen in the other placebo years (2011–2014), confirms that the 2015 estimate is an outlier driven by measurement constraints at the age 2.5 boundary.
  - Variance Compression: The standard deviation for the 2015 cohort is significantly lower (12.84) than all other cohorts (SD $\approx$ 18.5). This, combined with a higher minimum score (80 vs. 55 in other years), is symptomatic of a "floor effect". The test appears too difficult for 2 year old to provide meaningful variation at the bottom of the distribution, resulting in an artificially inflated mean and compressed variance that biases trend estimations.
  - The 2015 cohort has the smallest sample size ($N = 512$, other cohorts have 800 and 2000 obs depending if they were a refreshment sample or the original one respectively) since not all children of the 2015 cohort were able to take the test since the age restriction (older than 2.5). Therefore, this cohort is not representative of the 2015 cohort. Also, it poses individual observations disproportionate weight relative to the more stable treated cohorts.
- **Empirical Validation of Parallel Trends (2011–2014)**. Excluding the 2015 boundary case, the parallel trends assumption holds. From 2011 to 2014 (four placebo years), the treated and control municipalities follow an identical upward slope in raw cognitive scores. The gap between groups remains constant across these four cohorts, providing empirical support for the DiD. Furthermore, a joint F-test of the pre-treatment coefficients (2012–2015) is non-significant (p-value=0.25), suggesting that even with the 2015 noise, there is no statistically detectable systematic trend.
- **Sensitivity to 2015 cohort and to municipality linear trends**. Another compelling evidence that the estimates are similar with the 2015 cohort and the 2015 cohort introduces a spurious trend is found in the sensitivity analysis:
  - With 2015: Including municipality-specific linear trends doubles the treatment coefficients. This suggests that the model is attempting to correct for a trend that is actually just a measurement glitch in the earliest data point (see Tables 2 and 3).

– Without 2015: The DiD estimates become highly stable (see Table C.22). The coefficients with and without municipality linear trends are virtually identical. This stability demonstrates that once the psychometrically compromised 2015 cohort is removed, the identification is cleaner and the results are not driven by underlying regional trajectories.

As a conclusion for the Peabody analysis, the violation in 2015 is an artifact of the instrument's inability to distinguish performance at the age 2 lower bound. Given the identical slopes between 2011 and 2014 and the stability of the estimates upon the removal of 2015, we are confident that the earthquake's impact is accurately identified.

Table B.1: Descriptive Statistics of Peabody Scores by Birth Cohort 7 years later

Birth Cohort	**Status**	**Mean**	**SD**	**Min**	**Max**	**Sample (N)**
2015	Post-EQ (Outlier)	105.59	12.84	80	145	512
2014	Post-EQ (Placebo)	104.58	17.28	66	145	808
2013	Post-EQ (Placebo)	105.84	21.30	55	145	822
2012	Post-EQ (Placebo)	110.93	18.78	55	145	848
2011	Post-EQ (Placebo)	113.45	18.21	55	145	798
2010	Exposed (In utero)	113.08	18.00	55	145	808
2009	Exposed (Age 1)	116.18	18.58	55	145	2,042
2008	Exposed (Age 2)	116.97	17.78	55	145	2,516
2007	Exposed (Age 3)	117.45	18.47	55	145	2,528
2006	Exposed (Age 4)	116.94	17.91	55	145	2,506

Table B.2: Descriptive Statistics of Peabody Scores by Birth Cohort 2 years later

Birth Cohort	**Status**	**Mean**	**SD**	**Min**	**Max**	**Sample (N)**
2015	Post-EQ (Outlier)	105.59	12.84	80	145	512
2014	Post-EQ (Placebo)	104.58	17.28	66	145	808
2013	Post-EQ (Placebo)	105.84	21.30	55	145	822
2012	Post-EQ (Placebo)	110.93	18.78	55	145	848
2011	Post-EQ (Placebo)	113.45	18.21	55	145	798
2010	Exposed (In utero)	112.96	18.00	55	145	813
2009	Exposed (Age 1)	115.78	18.72	55	145	2,078
2008	Exposed (Age 2)	116.66	17.95	55	145	2,553
2007	Exposed (Age 3)	117.14	18.70	55	145	2,564
2006	Exposed (Age 4)	116.70	18.19	55	145	2,527

B.2 2011 as the Baseline

I have chosen the 2011 cohort as the baseline for the event cohort study. There may exist concerns since this cohort might be partially contaminated by the aftermath of the earthquake. The reasons to justify that it is a plausible choice are the following:

- **Statistical Robustness:** I re-estimated the main specification excluding the 2011 cohort. The results remain stable in both magnitude and statistical significance (see Table C.30 in the

Appendix and discussion in the robustness Section 4.3), confirming that the 2011 cohort does not bias the primary estimates. Additionally, I run the main specification adding the cohort 2011 interacted with the earthquake indicator. The results in Table C.15 shows a small and insignificant effect of the cohort 2011.

- **Event Study Evidence:** The event studies (Figures and 2 and 3) show that the 2012–2014 cohorts are statistically indistinguishable from the 2011 baseline. The absence of a recovery trend in the post-period indicates that outcomes had already reached a steady state by 2011, despite ongoing housing reconstruction.

- **Institutional Context:** While housing reconstruction was slow, essential human capital infrastructure recovered rapidly. Assessments by CEPAL 2010 indicate that 92.7% of health facilities suffered minor or no damage, with the majority becoming fully operational by June 2010 (Government of Chile 2010). Moreover, 44% of establishments with structural damage were concentrated in only one region (Maule region)[46]. The 2011 cohort thus benefited from a functional healthcare and educational environment, distinguishing them from the 2006–2010 cohorts who faced acute developmental shocks. Additionally, according to the Ministry of Education and the National Kindergartens Board by April and May 2010 schools and public day care resumed their operations respectively.

As a conclusion, the 2011 cohort should not be as contaminated as the 2006-10 cohorts and the choice of 2011 as a baseline seems to be plausible in this context. In any case, the estimates should be considered a lower bound of the true effects.

C Appendix: Tables

Table C.1: Age that cohorts take the tests

	Affected			Non-Affected		
	Age at Earthquake	Age Measured	Wave	Age at Earthquake	Age Measured	Wave
Panel A: Short-term						
Battelle	In utero to 4 yo	20 to 83 mo	2012	Unborn	30 to 83 mo	2017
Peabody	In utero to 4 yo	30 to 151 mo	2012	Unborn	30 to 91 mo	2017
CBCL1	In utero to 4 yo	20 to 71 mo	2012	Unborn	18 to 72 mo	2017
Panel B: Medium-term						
Peabody	In utero to 4 yo	84 to 151 mo	2017	Unborn	30 to 91 mo	2017
CBCL2	In utero to 4 yo	84 to 151 mo	2017	Unborn	73 to 91 mo	2017

Notes: Affected means that these cohorts were in utero up to four years old at the moment of the earthquake. These cohorts were born between 2006 and 2010. Non-affected means cohorts who were born after the earthquake. These cohorts depending on the test, were born between 2011 and 2016. "yo" means years old and "mo" months old.

[46] Regressions in Table C.29 excluding the Maule Region lead to the same results.

Table C.2: Classification of Analysis Groups by Birth Cohort and Survey-wave

Horizon		Group	
		Affected group (Cohort 2006–2010)	Control group (Cohort 2011–2015)
Short-term	Age at survey	Ages 2–6	Ages 2–6
	Data source	ELPI 2012	ELPI 2017
Medium-term	Age at survey	Ages 7–11	Ages 2–6
	Data source	ELPI 2017	ELPI 2017

Notes: This Table clarifies the distinction between the two analysis periods. The short-term analysis compares children measured approximately two years post-shock (using the 2012 wave for the affected group) against the control cohorts from the 2017 wave. The medium-term analysis compares both groups contemporaneously using the 2017 wave, seven years after the 2010 earthquake.

Table C.3: Summary of Shocks and Human Capital Impacts in Latin America

Paper	**Natural Disaster**	**Results**
Rosales-Rueda 2018	Rainfall in Ecuador (97-98)	-0.1 SD in Peabody scores (in utero)
Aguilar and Vicarelli 2022	Floods in México (97-99)	-0.19 SD in Peabody scores (in utero-3y)
Duque 2017	Conflicts in Colombia (99-07)	-0.18/-0.16 SD (Peabody/Math) (in utero-5y)
Hidrobo 2014	Economic Crisis Ecuador 99	-6% in Peabody to the mean (in utero-5 y)
Pazos et al. 2024	Rainfall Perú	-1 SD (at 5y); -0.87 SD (at 8y) (in utero-2 y)
Caruso and Miller 2015	Earthquake Perú 70	-0.5 years of schooling (in utero-2y)

Note: SD stands for Standard Deviations. All results are statistically significant. In the final parenthesis the age of the children studied.

Table C.4: Summary Statistis

	ELPI		CASEN Earthquake Panel		Vital Records	
	Mean	N	Mean	N	Mean	N
Mother and Household Characteristics						
Age	33.83	15,728	29.78	4,350	27.26	2,432,333
Years of Schooling	12.43	15,723	11.47	4,350	12.14	2,432,333
Married or cohabiting	0.61	15,827	0.71	4,350	0.68	2,432,333
Family Size	4.58	15,827	4.88	4,350		
Mother Previous Mental Health	0.03	15,827				
Father Previous Mental Health	0.11	15,827				
Sibling Previous Mental Health	0.12	15,827				
Children Characteristics						
Age (in months)	81.15	15,827				
Gender	0.51	15,827				
Birth Order	1.74	15,827				
Exposed to the Earthquake	0.45	15,827				

Notes: ELPI and CASEN Earthquake Panel include sampling weight at national level.

Table C.5: Balance Table by Gender

	(1)	(2)	(3)	(4)	(5)	(6)
	Boys			Girls		
	Non-Earthquake	Earthquake	Difference	Non-Earthquake	Earthquake	Difference
Children	2.024	2.071	0.047	1.995	2.075	0.080
	(0.955)	(0.969)	(0.040)	(0.986)	(0.959)	(0.062)
HH Size	4.499	4.577	0.078	4.473	4.626	0.154
	(1.637)	(1.533)	(0.122)	(1.649)	(1.581)	(0.141)
Father Present	0.587	0.638	0.052**	0.562	0.614	0.052*
	(0.492)	(0.481)	(0.025)	(0.496)	(0.487)	(0.028)
Mothers' Age	32.976	33.971	0.996***	33.231	34.109	0.878***
	(7.586)	(7.609)	(0.334)	(7.422)	(7.628)	(0.272)
Mothers' Schooling	12.347	12.484	0.138	12.302	12.445	0.142
	(2.775)	(3.090)	(0.203)	(2.913)	(3.136)	(0.196)
Fathers' Schooling	12.081	12.425	0.344	12.240	12.370	0.129
	(2.891)	(3.288)	(0.297)	(3.100)	(3.353)	(0.274)
Birth Order	1.729	1.730	0.001	1.708	1.761	0.053
	(0.882)	(0.859)	(0.033)	(0.880)	(0.888)	(0.049)
Income (2017)	238,779.797	257,967.500	19,187.709	251,531.250	256,453.141	4,921.898
	(209070.531)	(312357.656)	(19,775.055)	(213978.141)	(313799.500)	(20,735.896)
Observations	1,999	6,096	8,095	1,923	5,809	7,732

Notes: ELPI uses sampling weight at national level.

Table C.6: Selection on Observables: ELPI Survey

	# Children	HH size	Father Present	Mother age	Mother school	Father schooling	Birth order
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Affected*Earthquake	0.0255	-0.0429	-0.0165	-0.376	-0.301*	-0.383*	-0.0309
	(0.0467)	(0.0888)	(0.0252)	(0.454)	(0.181)	(0.212)	(0.0441)
Average	2.059	4.578	0.6148	33.83	12.43	12.34	1.739
Observations	15,789	15,789	15,789	15,789	15,789	10,115	15,789
$Adj - R^2$	0.0479	0.0649	0.0276	0.158	0.131	0.198	0.0180

Notes: Each column represents a separate regression. The dependent variable is the characteristic from the ELPI survey. It contains cohort of birth and municipality fixed effects. Standard errors clustered at the municipal level appear in parentheses—number of clusters. * p < 0.1, ** p < 0.05, *** p < 0.01.

Table C.7: Selection on Observables by Gender: ELPI Survey

	# Children	Father Present	Mother age	Mother schooling	Father schooling	Birth order
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Girl						
Affected*Earthquake	-0.0723	-0.0544	-0.503	-0.299	-0.438	-0.115**
	(0.0632)	(0.0378)	(0.562)	(0.244)	(0.289)	(0.0514)
Average	2.06	0.66	33.94	12.41	12.35	1.75
Observations	7,701	7,701	7,701	7,701	4,892	7,701
$Adj - R^2$	0.0641	0.0404	0.152	0.140	0.205	0.0403
Panel B: Boys						
Affected*Earthquake	0.102	0.0135	-0.0304	-0.297	-0.370	0.0475
	(0.0681)	(0.0322)	(0.569)	(0.220)	(0.273)	(0.0636)
Average	2.06	0.67	33.75	12.45	12.35	1.73
Observations	8,058	8,058	8,058	8,058	5,203	8,058
$Adj - R^2$	0.0538	0.0259	0.176	0.136	0.203	0.0129

Notes: Each column represents a separate regression. The dependent variable is the characteristic from the ELPI survey. It contains cohort of birth and municipality fixed effects. Standard errors clustered at the municipal level appear in parentheses—number of clusters. * p < 0.1, ** p < 0.05, *** p < 0.01.

Table C.8: Selection on Observables: Vital Birth Records

	# Children	Father Present	Mother age	Mother schooling	Father schooling	Birth order
	(1)	(2)	(3)	(4)	(5)	(6)
Affected*Earthquake	-0.0111*	0.00115	-0.120	-0.0806*	-0.0950*	-0.00647*
	(0.00638)	(0.00162)	(0.0733)	(0.0445)	(0.0512)	(0.00373)
Average	2.224	0.68	29.1	12.52	12.71	2.049
Observations	739,770	739,770	739,763	738,495	683,267	739,770
$Adj - R^2$	0.115	0.00886	0.0912	0.159	0.175	0.0362

Notes: Each column represents a separate regression. The dependent variable is the characteristic from the Vital Birth Records of Chile. Standard errors clustered at the municipal level appear in parentheses—number of clusters. It contains cohort of birth and municipality fixed effects.

Table C.9: Descriptive Statistics: ELPI vs. CASEN Earthquake Panel

Variable	ELPI	CASEN	p-value	Cohen's d
Family size	4.90	5.03	0.03	0.08
	(1.66)	(1.70)		
Mothers' schooling	11.47	11.20	0.04	0.18
	(3.26)	(3.17)		
Mothers' age	29.78	30.19	0.22	0.13
	(7.23)	(9.00)		
Income in 2010	101,435	119,952	0.01	0.05
	(194,666)	(113,137)		
Married or cohabitating	0.71	0.69	0.19	0.07
	(0.46)	(0.46)		
Fathers' Schooling	11.48	11.30	0.30	0.2
	(3.55)	(3.16)		
Observations	9,471	4,350		

Note: The table compares pre-earthquake characteristics between the ELPI and CASEN Earthquake Panel. ELPI children correspond to children who were conceived before the earthquake. Columns 1 and 2 report the weighted means for each survey, using their respective sampling weights to ensure population representativeness. The p value column shows the p value results of a weighted OLS regression, where each characteristic is regressed on a survey-source dummy; the reported p-values correspond to the test of the null hypothesis that the difference between the two weighted means is zero. Given that ELPI is the double in size, and this can yield statistical significance for negligible differences, we also report Cohen's d (Standardized Mean Difference) as a measure of effect size (Cohen 1988). Following standard literature Austin 2011, an absolute Cohen's d below 0.1 indicates a negligible imbalance, while values below 0.2 are considered "small" differences, suggesting that the two datasets are practically well-balanced for the analysis.

Table C.10: Correlation between PGA and Baseline Characteristics CASEN Earthquake Panel

	(1) # Children	(2) Mother age	(3) Mother educ	(4) Father Pres.	(5) HH size	(6) Income pp	(7) House Quality
PGA	-0.243*	-0.823	0.259	-0.129	-0.310	-29,506	-0.1387
	(0.147)	(1.473)	(0.655)	(0.0908)	(0.423)	(21,548)	(0.6137)
Observations	4,118	4,118	4,118	4,118	4,118	4,118	4,118
$Adj - R^2$	0.0012	0.0000	0.0001	0.0013	0.0004	0.0012	0.0001

Note: Each column represents a separate OLS regression. *PGA* denotes Peak Ground Acceleration (m/s^2) and the data goes from zero to a 0.54. House Quality is a categorical variable defined as: (1) Acceptable, (2) Salvageable (recoverable), and (3) Beyond repair (unsalvageable). Regression from Column7 is a ordered logit. Father Pres. is a dummy that takes the value one if the father is present in the household. Income pp is the monthly incomer per capita. All models include sampling weights. Robust standard errors (in parentheses) are clustered at the municipality level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.11: DiD in Child Daycare or School Enrollment

	(1) 0-4 years-old	(2) Selected child	(3) 5-18 years-old yo	(4) Non-selected child
Earthquake*Post	-0.0339	0.0164	-0.0015	0.0163
	(0.0313)	(0.0132)	(0.0111)	(0.0104)
Average	0.3757	43.19	0.9258	0.8264
Source	CASEN	ELPI	CASEN	ELPI
Time later	3 months	2 years	3 months	2 years
Observations	11,396	23,384	31,030	28,698
$Adj - R^2$	0.532	0.1962	0.316	0.2042

Notes: Estimates from a Difference-in-Differences (DiD) specification at the household level. The dependent variable is a binary indicator for school or daycare enrollment for children aged 0–4 or 5–18 in the case of CASEN Earthquake Panel. Selected or interviewed children from ELPI are between 0 to 4 years-old and non-selected children could be any age less than 18 years-old. Each column reports a separate regression. The treatment group consists of households with at least one child under age four. Post is an indicator that takes the value 1 if the period measured is after the earthquake. All models include controls for household size and income, number of children, mother's age and education, and father's presence. Municipality and year fixed effects are included in all specifications. Standard errors clustered at the municipality level (145 clusters) are in parentheses. Estimates are weighted using national-level sampling weights. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.12: Within-Region Variation in Peak Ground Acceleration (PGA)

Region	Mean PGA	Std. Dev.	Min PGA	Max PGA
1	0.0020	0.0000	0.002	0.002
2	0.0045	0.0007	0.003	0.005
3	0.0131	0.0028	0.009	0.019
4	0.0480	0.0090	0.033	0.124
5	0.2747	0.0715	0.193	0.453
6	0.3498	0.0177	0.302	0.477
7	0.3650	0.0188	0.329	0.486
8	0.4348	0.0637	0.257	0.531
9	0.3228	0.0586	0.244	0.426
10	0.0494	0.0184	0.023	0.074
11	0.0099	0.0011	0.005	0.011
12	0.0010	0.0000	0.001	0.001
13	0.3061	0.0300	0.182	0.401
14	0.1121	0.0075	0.071	0.117
15	0.0020	0.0000	0.002	0.002

Notes: The table reports the mean, standard deviation, minimum, and maximum values of peak ground acceleration (PGA) at the municipality level within each region. PGA is measured in g. Substantial within-region variation is observed in the most affected regions, indicating heterogeneity in seismic exposure across municipalities.

Table C.13: Fatalities by Region

Region	Fatalities	Average PGA
Valparaíso	25	0.25
O'Higgins	53	0.36
Maule	280	0.38
Biobío	145	0.41
Araucanía	17	0.26
Metropolitana	27	0.30
All regions affected	**547**	0.33

Note: Larrañaga and Herrera 2010 and final official figures from the Ministry of the Interior and Public Security (January 2011). While initial reports in March 2010 estimated over 800 casualties, the official total was corrected downward following victim identification. The figures in this table reflect verified fatalities; some databases report a total of 550 to include 25 missing persons.

Table C.14: The Impact of the Earthquake on Cognitive and Non-Cognitive Development: Placebo Estimates in Children born after the Earthquake

	(1) Battelle	(2) Peabody	(3) CBCL1	(4) CBCL2
Half Born After*Earthquake	0.0750	0.0824	-0.0440	0.103
	(0.0797)	(0.0893)	(0.0920)	(0.145)
Observations	4,384	3,760	4,098	1,103
$Adj - R^2$	0.193	0.199	0.195	0.106

Each column represents a separate regression. The dependent variable is the test score measured in standard deviations. Peabody is the Peabody Test Score, Battelle is the Battelle Test Score and CBCL is the Child Behavior Check List measure. These estimates are done using the children born after the earthquake. I define as Affected the first half of of children born after the earthquake. Observations are at the individual/municipality level. Standard errors clustered at the municipal level appear in parentheses—number of clusters: 145. The estimates use sampling weights representative at the national level. Child/Mother/HH characteristics include gender, age at the moment of the test (in months), birth order, mothers' age and schooling, father present, number member of the household, prior mental health condition for mother, father and relative. All specifications contain cohort of birth and municipality fixed effects. * p < 0.1, ** p < 0.05, *** p < 0.01.

Table C.15: The Impact of the Earthquake on Cognitive and Non-Cognitive Development: adding placebo 2011 cohort

	(1)	(2) Short-Term	(3)	(4) Medium-Term	(5)
	Battelle	Peabody	CBCL1	Peabody	CBCL2
Affected*Earthquake	-0.0793	-0.126	0.136	-0.164**	-0.0714
	(0.0962)	(0.0873)	(0.105)	(0.0704)	(0.236)
Cohort2011*Earthquake	0.0152	0.0655	0.214	0.0465	0.0737
	(0.102)	(0.0919)	(0.174)	(0.0908)	(0.275)
Observations	13,070	14,205	11,654	14,069	11,568
$Adj - R^2$	0.224	0.142	0.137	0.185	0.107

Notes: Each column represents a separate regression. The dependent variable is the test score measured in standard deviations. Peabody is the Peabody Test Score, Battelle is the Battelle Test Score and CBCL is the Child Behavior Check List measure. I exclude the first cohort born after the earthquake. Observations are at the individual/municipality level. Short- and Medium-term mean measured two and seven years later. Standard errors clustered at the municipal level appear in parentheses—number of clusters: 145. The estimates use sampling weights representative at the national level. Child/Mother/HH characteristics include gender, age at the moment of the test (in months), birth order, mothers' age and schooling, father present, number member of the household, prior mental health condition for mother, father and relative. Pre-control characteristics are covariates measured previous to the earthquake and includes household income per capita, mother works indicator, number of siblings and a dummy for missing pre-control characteristics for children who where born after the first wave of the survey. All specifications contain cohort of birth, municipality fixed effects, and municipality linear trends. * p < 0.1, ** p < 0.05, *** p < 0.01.

Table C.16: The Impact of the Earthquake on Cognitive and Non-Cognitive Development on Unaffected Municipalities using PGA as exposure variable

	(1)	(2)	(3)	(4)	(5)
		Short-term		Medium-term	
	Battelle	Peabody	CBCL1	Peabody	CBCL2
Affected*PGA	-0.0135	-0.00695	0.0278	-0.00891	0.00578
	(0.0171)	(0.0188)	(0.0190)	(0.0151)	(0.0156)
Observations	3,262	3,510	2,865	3,453	2,980
$Adj - R^2$	0.191	0.149	0.149	0.198	0.131

Notes: Each column represents a separate regression. The dependent variable is the test score measured in standard deviations. Peabody is the Peabody Test Score, Battelle is the Battelle Test Score and CBCL is the Child Behavior Check List measure. I exclude the Metropolitan, V, VI, VII, VII and IX regions. Observations are at the individual/municipality level. Short- and Medium-term mean measured two and seven years later. Standard errors clustered at the municipal level appear in parentheses—number of clusters: 145. The estimates use sampling weights representative at the national level. Child/Mother/HH characteristics include gender, age at the moment of the test (in months), birth order, mothers' age and schooling, father present, number member of the household, prior mental health condition for mother, father and relative. All specifications contain cohort of birth and municipality fixed effects. * p < 0.1, ** p < 0.05, *** p < 0.01.

Table C.17: The Impact of the Earthquake on Cognitive and Non-Cognitive Development in Unaffected Municipalities using a North Regions as the affected ones

	(1)	(2)	(3)	(4)	(5)
		Short-term		Medium-term	
	Battelle	Peabody	CBCL	Peabody	CBCL
Affected*North	0.235	0.209	0.0450	0.145	0.0808
	(0.150)	(0.171)	(0.223)	(0.136)	(0.142)
Observations	3,262	3,510	2,865	3,453	2,980
$Adj - R^2$	0.196	0.151	0.147	0.199	0.131

Notes: Each column represents a separate regression. The dependent variable is the test score measured in standard deviations. Peabody is the Peabody Test Score, Battelle is the Battelle Test Score and CBCL is the Child Behavior Check List measure. I exclude the Metropolitan, V, VI, VII, VII and IX regions and North is a indicator value that takes the value 1 for the XV, I, II, III and IV region and zero otherwise. Observations are at the individual/municipality level. Short- and Medium-term mean measured two and seven years later. Standard errors clustered at the municipal level appear in parentheses—number of clusters: 145. The estimates use sampling weights representative at the national level. Child/Mother/HH characteristics include gender, age at the moment of the test (in months), birth order, mothers' age and schooling, father present, number member of the household, prior mental health condition for mother, father and relative. All specifications contain cohort of birth and municipality fixed effects. * p < 0.1, ** p < 0.05, *** p < 0.01.

Table C.18: Migration using Mother Fixed Effects and Vital Records

	(1) Municipality	(2) Region	(3) NA to A	(4) A to NA
Affected*Earthquake	0.000245	-0.00521***	0.000791***	-0.00393***
	(0.00162)	(0.000687)	(0.000130)	(0.000360)
Average	0.2494	0.0727	0.0117	0.0146
Observations	738,496	738,496	738,496	738,496
$Adj - R^2$	0.0400	0.0329	0.00732	0.0641

Notes: Each column represents a separate regression. These estimates use data from the Vital Birth Records and a mother fixed effect model. That means that only mother who had at least two children over 2005 and 2015 show up in these estimates. The dependent variable in Column 1 is a dummy for municipality migration, Column2 for region migration, Column3 for migration from a non-affected municipality (NA) to an affected one and Column4 for migration from a, affected municipality to a non-affected one. Child/Mother/HH characteristics include mothers' age and schooling, father present, number of children of the household, urban, and mother single status. Standard errors clustered at the mother level appear in parentheses. It contains cohort of birth fixed effects. * p < 0.1, ** p < 0.05, *** p < 0.01.

Table C.19: Fertility Decisions

	(1) # Births	(2) Planned	(3) Male/Female	(4) Male/Female
Affected*Earthquake	0.465	-0.008	0.001	-0.019
	(0.988)	(0.031)	(0.003)	(0.022)
Average	60.58	0.4565	0.5094	0.5112
Data Source	Vital	ELPI	Vital	ELPI
Observations	49,994	14,545	51,704	8,806
$Adj - R^2$	0.989	0.126	0.003	0.006

Notes: Each column represents a separate regression. The dependent variable in Column 1 is the number of birth at municipality/month level and exploits year and geographic variation across municipalities in the number of births. the dependent variable in Column2 is a dummy that takes the value one if the birth was planned and observations are at the individual/municipality level and exploits cohort and geographic variation. Finally, Columns 3 and 4 are the ratio of male to females births and exploit municipality and year variation. Standard errors clustered at the municipal level appear in parentheses—number of clusters. Columns 2 and 4 use sampling weights representative at the national level. Column 2 controls for Child/Mother/HH characteristics including gender, age at the moment of the test (in months), birth order, mothers' age and schooling, father present, number member of the household, prior mental health condition for mother, father and relative. All columns control for birth cohort and municipality fixed effects. * p < 0.1, ** p < 0.05, *** p < 0.01.

Table C.20: Attrition Rate

	(1) Missing(=1)	(2) Missing(=1)
Affected*Earthquake	0.0609**	0.0596
	(0.0308)	(0.0378)
Mun. Linear trends	×	✓
Observations	18,129	18,129
$Adj - R^2$	0.0659	0.0664

Notes: Each column represents a separate regression. The dependent variable in Columns 1 and 2 is a indicator variable that takes the value one if the observation was missed in wave 3, zero otherwise. Standard errors clustered at the municipal level appear in parentheses—number of clusters. It contains cohort of birth and municipality fixed effects. * p < 0.1, ** p < 0.05, *** p < 0.01.

Table C.21: The Impact of the Earthquake on Cognitive and Non-Cognitive Development: Kling and Liebman (2004) and Lee (2009) bounds

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
		Lower bounds		Coefficient		Upper Bounds	
	Lee (-)	(+/-) .25 SD	(+/-) .1 SD		(-/+) .1 SD	(-/+) .25 SD	Lee(+)
Panel A: Medium-Term							
Peabody	-0.137**	-0.102	-0.175**	-0.216***	-0.272***	-0.346***	-0.291***
	(0.0672)	(0.0729)	(0.0738)	(0.0721)	(0.0751)	(0.0763)	(0.0738)
CBCL2	-0.127*	-0.0235	-0.117*	-0.137*	-0.190***	-0.261***	-0.172***
	(0.0760)	(0.0705)	(0.0709)	(0.0734)	(0.0716)	(0.0723)	(0.0631)
Panel B: Short-Term							
Battelle	-0.0985	-0.0563	-0.107	-0.175**	-0.244***	-0.324***	-0.220***
	(0.0785)	(0.0779)	(0.0780)	(0.0799)	(0.0783)	(0.0787)	(0.0796)
Peabody	-0.149*	-0.103	-0.173**	-0.209**	-0.266***	-0.335***	-0.297***
	(0.0892)	(0.0862)	(0.0869)	(0.0893)	(0.0881)	(0.0891)	(0.0802)
CBCL1	0.215**	0.255***	0.166*	0.133	0.0473	-0.0419	0.0314
	(0.0871)	(0.0888)	(0.0876)	(0.0878)	(0.0861)	(0.0851)	(0.0934)

Notes: Each column represents a separate regression. The dependent variable is the test score measured in standard deviations. Column 4 presents the main estimates without sampling weights. Columns (1) and (7) are estimated by trimming the top/bottom of the treatment group by the difference in attrition between the treatment and control, as in Lee 2009; Columns (2) to (3) and (5) to (6) replace outcome values for the attritors with +/- 0.x standard deviations of their respective treatment group-wave means as in Kling et al. 2007, lower bounds subtract this value from the treatment group and add it to the control group and vice versa for the upper bounds. Peabody is the Peabody Test Score, Battelle is the Battelle Test Score and CBCL is the Child Behavior Check List measure. Observations are at the individual/municipality level. Short- and Medium-term mean measured two and seven years later. Standard errors clustered at the municipal level appear in parentheses—number of clusters: 145. Child/Mother/HH characteristics include gender, age at the moment of the test (in months), birth order, mothers' age and schooling, father present, number member of the household, prior mental health condition for mother, father and relative. All specifications contain cohort of birth and municipality fixed effects. * p < 0.1, ** p < 0.05, *** p < 0.01.

Table C.22: The Impact of the Earthquake on Cognitive and Non-Cognitive Development: No 2015 cohort

	(1)	(2)	(3)	(4)	(5)
		Short-Term			Medium-Term
	Battelle	Peabody	CBCL1	Peabody	CBCL2
Panel A:					
Non-Municipality Linear Trends					
Affected*Earthquake	-0.120	-0.187**	0.147	-0.218***	-0.129*
	(0.0975)	(0.0896)	(0.107)	(0.0718)	(0.0763)
Panel B:					
With Municipality Linear Trends					
Affected*Earthquake	-0.158	-0.238**	-0.0342	-0.222**	-0.220**
	(0.152)	(0.107)	(0.143)	(0.0853)	(0.105)
Observations	11,592	13,699	10,162	13,563	11,568

Notes: Each column represents a separate regression. The dependent variable is the test score measured in standard deviations. Peabody is the Peabody Test Score, Battelle is the Battelle Test Score and CBCL is the Child Behavior Check List measure. All estimates use cohort 2006-14. Observations are at the individual/municipality level. Short- and Medium-term mean measured two and seven years later. Standard errors clustered at the municipal level appear in parentheses—number of clusters: 145. The estimates use sampling weights representative at the national level. Child/Mother/HH characteristics include gender, age at the moment of the test (in months), birth order, mothers' age and schooling, father present, number member of the household, prior mental health condition for mother, father and relative. All specifications contain cohort of birth, and municipality fixed effects. * p < 0.1, ** p < 0.05, *** p < 0.01.

Table C.23: The Impact of the Earthquake on Cognitive and Non-Cognitive Development using Peak Ground Acceleration as Intensity Measure

	(1)	(2)	(3)	(4)	(5)
		Short-Term			Medium-Term
	Batelle	Peabody	CBCL1	Peabody	CBCL2
Affected*PGA	-0.00185	-0.00541**	0.00274	-0.00618***	-0.00412*
	(0.00328)	(0.00267)	(0.00289)	(0.00217)	(0.00232)
Observations	13,070	14,205	11,654	14,069	11,568
$Adj - R^2$	0.224	0.143	0.139	0.185	0.107

Notes: Each column represents a separate regression. The dependent variable is the test score measured in standard deviations. Peabody is the Peabody Test Score, Battelle is the Battelle Test Score and CBCL is the Child Behavior Check List measure. *PGA* denotes Peak Ground Acceleration (m/s^2) and the data goes fro zero to a 0.53. Observations are at the individual/municipality level. Short- and Medium-term mean measured two and seven years later. Standard errors clustered at the municipal level appear in parentheses—number of clusters: 145. The estimates use sampling weights representative at the national level. Child/Mother/HH characteristics include gender, age at the moment of the test (in months), birth order, mothers' age and schooling, father present, number member of the household, prior mental health condition for mother, father and relative. All specifications contain cohort of birth, and municipality fixed effects. * p < 0.1, ** p < 0.05, *** p < 0.01.

Table C.24: The Impact of the Earthquake on Cognitive and Non-Cognitive Development: Distance to the Epicenter (in kms)

	(1)	(2)	(3)	(4)	(5)
		Short-Term		Medium-Term	
	Battelle	Peabody	CBCL1	Peabody	CBCL2
Affected*Distance	0.000109	0.000135*	-0.000054	0.000138**	.0001046*
	(9.13e-05)	(7.89e-05)	(8.15e-05)	(6.15e-05)	(6.03e-05)
Observations	13,070	14,205	11,654	14,069	11,568
$Adj - R^2$	0.224	0.134	0.138	0.184	0.107

Notes: Each column represents a separate regression. The dependent variable is the test score measured in standard deviations. Peabody is the Peabody Test Score, Battelle is the Battelle Test Score and CBCL is the Child Behavior Check List measure. Distance is measured in kilometres. Observations are at the individual/municipality level. Short- and Medium-term mean measured two and seven years later. Standard errors clustered at the municipality level appear in parentheses—number of clusters: 145. In square brackets p values associated to wild cluster bootstrap. Child/Mother/HH characteristics include gender, age at the moment of the test (in months), birth order, mothers' age and schooling, father present, number member of the household, prior mental health condition for mother, father and relative. All specifications contain cohort of birth and municipality fixed effects. * p < 0.1, ** p < 0.05, *** p < 0.01.

Table C.25: The Impact of the Earthquake on Cognitive and Non-Cognitive Development: Coastal Municipalities

	(1)	(2)	(3)	(4)	(5)
		Short-Term		Medium-Term	
	Battelle	Peabody	CBCL1	Peabody	CBCL2
Affected*Earthquake*Coastal	0.0250	-0.0754	0.322	0.0664	0.0400
	(0.216)	(0.206)	(0.247)	(0.188)	(0.176)
Observations	13,070	14,205	11,654	14,069	11,568
$Adj - R^2$	0.224	0.134	0.138	0.185	0.107

Notes: Each column represents a separate regression. The dependent variable is the test score measured in standard deviations. Peabody is the Peabody Test Score, Battelle is the Battelle Test Score and CBCL is the Child Behavior Check List measure. Costal is a variable that identify coastal municipalities. Observations are at the individual/municipality level. Short- and Medium-term mean measured two and seven years later. Standard errors clustered at the municipal level appear in parentheses—number of clusters: 145. Child/Mother/HH characteristics include gender, age at the moment of the test (in months), birth order, mothers' age and schooling, father present, number member of the household, prior mental health condition for mother, father and relative. All specifications contain cohort of birth, municipality fixed effects, coastal municipality dummy and their interactions. * p < 0.1, ** p < 0.05, *** p < 0.01.

Table C.26: The Impact of the Earthquake on Cognitive and Non-Cognitive Development: Different Cutoffs

	(1)	(2) Short-Term	(3)	(4) Medium-Term	(5)
	Battelle	Peabody	CBCL1	Peabody	CBCL2
Panel A: $PGA \geq 0.18$					
Affected*Earthquake	-0.0802	-0.142*	0.126	-0.175**	-0.129*
	(0.0942)	(0.0873)	(0.100)	(0.0684)	(0.0763)
Panel B: $PGA \geq 0.3$					
Affected*Earthquake	-0.111	-0.113*	0.192**	-0.112**	-0.129*
	(0.0855)	(0.0665)	(0.0809)	(0.0548)	(0.0725)
Panel C: $PGA \geq 0.35$					
Affected*Earthquake	0.0469	-0.200***	0.147	-0.125**	-0.187**
	(0.142)	(0.0753)	(0.102)	(0.0595)	(0.0813)

Notes: Each column represents a separate regression. The dependent variable is the test score measured in standard deviations. Peabody is the Peabody Test Score, Battelle is the Battelle Test Score and CBCL is the Child Behavior Check List measure. Observations are at the individual/municipality level. Short- and Medium-term mean measured two and seven years later. Standard errors clustered at the municipality level appear in parentheses—number of clusters: 145. In square brackets p values associated to wild cluster bootstrap. Child/Mother/HH characteristics include gender, age at the moment of the test (in months), birth order, mothers' age and schooling, father present, number member of the household, prior mental health condition for mother, father and relative. All specifications contain cohort of birth and municipality fixed effects. * p < 0.1, ** p < 0.05, *** p < 0.01.

Table C.27: The Impact of the Earthquake on Cognitive and Non-Cognitive Development: Regional cluster and FE

	(1)	(2) Short-Term	(3)	(4) Medium-Term	(5)
	Battelle	Peabody	CBCL1	Peabody	CBCL2
Affected*Earthquake	-0.0791	-0.143*	0.135	-0.172***	-0.135*
	(0.112)	(0.0799)	(0.111)	(0.0570)	(0.0671)
	[0.52]	[0.07]	[0.32]	[0.00]	[0.04]
Observations	14,104	11,592	13,107	14,239	11,693
$Adj - R^2$	0.150	0.0597	0.181	0.121	0.0945

Notes: Each column represents a separate regression. The dependent variable is the test score measured in standard deviations. Peabody is the Peabody Test Score, Battelle is the Battelle Test Score and CBCL is the Child Behavior Check List measure. Observations are at the individual/municipality level. Short- and Medium-term mean measured two and seven years later. Standard errors clustered at the region level appear in parentheses—number of clusters: 145. In square brackets p values associated to wild cluster bootstrap. Child/Mother/HH characteristics include gender, age at the moment of the test (in months), birth order, mothers' age and schooling, father present, number member of the household, prior mental health condition for mother, father and relative. All specifications contain cohort of birth and region fixed effects. * p < 0.1, ** p < 0.05, *** p < 0.01.

Table C.28: The Impact of the Earthquake on Cognitive and Non-Cognitive Development Changing the Control Group

	(1)	(2)	(3)	(4)	(5)
		Short-term		Medium-term	
	Battelle	Peabody	CBCL1	Peabody	CBCL2
Panel A: Nearest Municipalities					
Affected*Earthquake	-0.0677	-0.136	0.146	-0.187*	-0.214**
	(0.126)	(0.113)	(0.145)	(0.0949)	(0.0970)
Observations	11,682	12,720	10,422	12,581	10,326
$Adj - R^2$	0.228	0.140	0.140	0.190	0.109
Panel B: Farthest Municipalities					
Affected*Earthquake	-0.192*	-0.150	0.127	-0.171**	-0.116
	(0.115)	(0.123)	(0.0894)	(0.0774)	(0.103)
Observations	11,470	12,180	10,255	12,007	10,065
$Adj - R^2$	0.229	0.143	0.128	0.192	0.0966

Notes: Each column represents a separate regression. The dependent variable is the test score measured in standard deviations. Peabody is the Peabody Test Score, Battelle is the Battelle Test Score and CBCL is the Child Behavior Check List measure. Nearest are the nearest municipalities in the IV, X, XVI regions. Furthest are the furthest municipalities in the regions I, II, III, XV, XI, XII. Observations are at the individual/municipality level. Short- and Medium-term mean measured two and seven years later. Standard errors clustered at the municipal level appear in parentheses—number of clusters: 145. The estimates use sampling weights representative at the national level. Child/Mother/HH characteristics include gender, age at the moment of the test (in months), birth order, mothers' age and schooling, father present, number member of the household, prior mental health condition for mother, father and relative. All specifications contain cohort of birth and municipality fixed effects. * p < 0.1, ** p < 0.05, *** p < 0.01.

Table C.29: The Impact of the Earthquake on Cognitive and Non-Cognitive Development: No Maule and Biobío Regions

	(1)	(2)	(3)	(4)	(5)
		Short-Term		Medium-Term	
	Batelle	Peabody	CBCL1	Peabody	CBCL2
Panel A: No Maule Region					
Affected*Earthquake	-0.0812	-0.148*	0.125	-0.173**	-0.123*
	(0.0964)	(0.0878)	(0.101)	(0.0693)	(0.0749)
Panel B: No Biobío Region					
Affected*Earthquake	-0.0928	-0.118	0.0605	-0.165**	-0.129*
	(0.0951)	(0.0879)	(0.100)	(0.0693)	(0.0775)
Panel C: Non Both from Above					
Affected*Earthquake	-0.0939	-0.105	0.0520	-0.163**	-0.122
	(0.0975)	(0.0890)	(0.100)	(0.0702)	(0.0819)

Notes: Each column represents a separate regression. The dependent variable is the test score measured in standard deviations. Peabody is the Peabody Test Score, Battelle is the Battelle Test Score and CBCL is the Child Behavior Check List measure. Distance is measured in kilometres. Observations are at the individual/municipality level. Short- and Medium-term mean measured two and seven years later. Standard errors clustered at the municipality level appear in parentheses—number of clusters: 145. In square brackets p values associated to wild cluster bootstrap. Child/Mother/HH characteristics include gender, age at the moment of the test (in months), birth order, mothers' age and schooling, father present, number member of the household, prior mental health condition for mother, father and relative. All specifications contain cohort of birth and municipality fixed effects. * p < 0.1, ** p < 0.05, *** p < 0.01.

Table C.30: The Impact of the Earthquake on Cognitive and Non-Cognitive Development dropping the first year cohort born after the shock

	(1)	(2)	(3)	(4)	(5)
	Short-term			Medium-term	
	Battelle	Peabody	CBCL1	Peabody	CBCL2
Affected*Earthquake	-0.0812	-0.123	0.142	-0.162**	0.129
	(0.0964)	(0.0870)	(0.106)	(0.0699)	(0.243)
Observations	12,489	13,349	11,413	13,213	10,702
$Adj - R^2$	0.229	0.140	0.141	0.187	0.109

Notes: Each column represents a separate regression. The dependent variable is the test score measured in standard deviations. Peabody is the Peabody Test Score, Battelle is the Battelle Test Score and CBCL is the Child Behavior Check List measure. I exclude the first cohort born after the earthquake. Observations are at the individual/municipality level. Short- and Medium-term mean measured two and seven years later. Standard errors clustered at the municipal level appear in parentheses—number of clusters: 145. The estimates use sampling weights representative at the national level. Child/Mother/HH characteristics include gender, age at the moment of the test (in months), birth order, mothers' age and schooling, father present, number member of the household, prior mental health condition for mother, father and relative. All specifications contain cohort of birth, municipality fixed effects, and municipality linear trends. * p < 0.1, ** p < 0.05, *** p < 0.01.

Table C.31: The Impact of the Earthquake on Cognitive and Non-Cognitive Development in the Medium-Term in smaller windows

	(1)	(2)	(3)	(4)	(5)	(6)
	Peabody			CBCL		
Affected*Earthquake	-0.249**	-0.205**	-0.240***	-0.200*	-0.174**	-0.153*
	(0.111)	(0.0785)	(0.0745)	(0.118)	(0.0856)	(0.0818)
Cohorts	2010-11	2009-12	2008-2012	2010-11	2009-12	2008-2011
Observations	1,586	4,456	7,756	1,615	3,942	6,490
$Adj - R^2$	0.143	0.156	0.183	0.116	0.121	0.115

Notes: Each column represents a separate regression. The dependent variable is the test score measured in standard deviations. Peabody is the Peabody Test Score, Battelle is the Battelle Test Score and CBCL is the Child Behavior Check List measure. I exclude the first cohort born after the earthquake. Observations are at the individual/municipality level. Short- and Medium-term mean measured two and seven years later. Standard errors clustered at the municipal level appear in parentheses—number of clusters: 145. The estimates use sampling weights representative at the national level. Child/Mother/HH characteristics include gender, age at the moment of the test (in months), birth order, mothers' age and schooling, father present, number member of the household, prior mental health condition for mother, father and relative. All specifications contain cohort of birth, municipality fixed effects, and municipality linear trends. * p < 0.1, ** p < 0.05, *** p < 0.01.

Table C.32: The Impact of the Earthquake on Cognitive and Non-Cognitive Development without Sampling Weights

	(1)	(2)	(3)	(4)	(5)
		Short-term		Medium-term	
	Battelle	Peabody	CBCL1	Peabody	CBCL2
Affected*Earthquake	-0.175**	-0.209**	0.133	-0.216***	-0.137*
	(0.0799)	(0.0893)	(0.0878)	(0.0721)	(0.0734)
Observations	13,695	14,792	12,224	14,069	11,658
$Adj - R^2$	0.240	0.116	0.115	0.152	0.147

Notes: Each column represents a separate regression. The dependent variable is the test score measured in standard deviations. Peabody is the Peabody Test Score, Battelle is the Battelle Test Score and CBCL is the Child Behavior Check List measure. Observations are at the individual/municipality level. Short- and Medium-term mean measured two and seven years later. Standard errors clustered at the municipal level appear in parentheses—number of clusters: 145. Child/Mother/HH characteristics include gender, age at the moment of the test (in months), birth order, mothers' age and schooling, father present, number member of the household, prior mental health condition for mother, father and relative. All specifications contain cohort of birth and municipality fixed effects. * p < 0.1, ** p < 0.05, *** p < 0.01.

Table C.33: The Impact of the Earthquake on Cognitive and Non-Cognitive Development: Multiple Hypothesis Testing

	(1)	(2)	(3)	(4)	(5)
		Short-term		Medium-term	
	Battelle	Peabody	CBCL1	Peabody	CBCL2
Affected*Earthquake	-0.0813	-0.141*	0.126	-0.174**	-0.129*
	(0.0944)	(0.0855)	(0.100)	(0.0686)	(0.0763)
	[0.26]	[0.12]	[0.22]	[0.02]	[0.10]

Notes: Each column represents a separate regression. The dependent variable is the test score measured in standard deviations. In square brackets are Romano-Wolf standard erores from 500 replications. Peabody is the Peabody Test Score, Battelle is the Battelle Test Score and CBCL is the Child Behavior Check List measure. Observations are at the individual/municipality level. Short- and Medium-term mean measured two and seven years later. Standard errors clustered at the municipal level appear in parentheses—number of clusters: 145. Child/Mother/HH characteristics include gender, age at the moment of the test (in months), birth order, mothers' age and schooling, father present, number member of the household, prior mental health condition for mother, father and relative. All specifications contain cohort of birth and municipality fixed effects. * p < 0.1, ** p < 0.05, *** p < 0.01.

Table C.34: The Impact of the Earthquake on Cognitive and Non-Cognitive Development by Mothers' Education

	(1)	(2)	(3)	(4)	(5)
		Short-term			Medium-term
	Battelle	Peabody	CBCL1	Peabody	CBCL2
Panel A: Less than HS					
Affected*Earthquake	-0.141	-0.149	0.166	-0.172**	-0.143
	(0.0946)	(0.0982)	(0.132)	(0.0736)	(0.111)
Observations	8,777	9,657	7,794	9,556	8,127
$Adj - R^2$	0.221	0.125	0.135	0.167	0.107
Panel B: More than HS					
Affected*Earthquake	-0.00725	-0.147	0.0487	-0.204*	-0.113
	(0.117)	(0.116)	(0.0926)	(0.120)	(0.120)
Observations	4,270	4,526	3,835	4,493	3,426
$Adj - R^2$	0.231	0.125	0.127	0.177	0.0873
p-value LessHS=MoreHS	0.724	0.685	0.519	0.572	0.901

Notes: Each column represents a separate regression. The dependent variable is the test score measured in standard deviations. Peabody is the Peabody Test Score, Battelle is the Battelle Test Score and CBCL is the Child Behavior Check List measure. Observations are at the individual/municipality level. Short- and Medium-term mean measured two and seven years later. Standard errors clustered at the municipal level appear in parentheses—number of clusters: 145. The estimates use sampling weights representative at the national level. Child/Mother/HH characteristics include gender, age at the moment of the test (in months), birth order, mothers' age and schooling, father present, number member of the household, prior mental health condition for mother, father and relative. All specifications contain cohort of birth and municipality fixed effects. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table C.35: The Impact of the Earthquake on Cognitive and Non-Cognitive Development by Mothers' Age

	(1)	(2)	(3)	(4)	(5)
		Short-Term		Medium-Term	
	Battelle	Peabody	CBCL1	Peabody	CBCL2
Panel A: Mothers' age≤34					
Affected*Earthquake	-0.164	-0.199**	0.121	-0.213***	-0.217**
	(0.109)	(0.0958)	(0.106)	(0.0802)	(0.0856)
Observations	6,482	6,768	5,951	6,717	5,068
$Adj - R^2$	0.223	0.153	0.143	0.191	0.114
Panel B: Mothers' age>34					
Affected*Earthquake	-0.0278	-0.0798	0.0828	-0.126	-0.0453
	(0.106)	(0.100)	(0.133)	(0.0892)	(0.120)
Observations	6,568	7,411	5,678	7,326	6,480
$Adj - R^2$	0.247	0.128	0.136	0.161	0.109
P value A=B	0.075	0.09	0.903	0.279	0.37

Notes: Each column represents a separate regression. The dependent variable is the test score measured in standard deviations. Peabody is the Peabody Test Score, Battelle is the Battelle Test Score and CBCL is the Child Behavior Check List measure. 34 years old is the median age of mothers in the sample. Observations are at the individual/municipality level. Short- and Medium-term mean measured two and seven years later. Standard errors clustered at the municipal level appear in parentheses—number of clusters: 145. The estimates use sampling weights representative at the national level. Child/Mother/HH characteristics include gender, age at the moment of the test (in months), birth order, mothers' age and schooling, father present, number member of the household, prior mental health condition for mother, father and relative. All specifications contain cohort of birth and municipality fixed effects. * p < 0.1, ** p < 0.05, *** p < 0.01.

Table C.36: The Impact of the Earthquake on Cognitive and Non-Cognitive Development by Father Present

	(1)	(2)	(3)	(4)	(5)
		Short-Term			Medium-Term
	Battelle	Peabody	CBCL1	Peabody	CBCL2
Panel A: Father Present=1					
Affected*Earthquake	-0.166	-0.169	0.233*	-0.161**	-0.179
	(0.101)	(0.107)	(0.134)	(0.0751)	(0.119)
Observations	5,058	5,555	4,518	5,495	4,560
$Adj - R^2$	0.218	0.145	0.148	0.186	0.140
Panel B: Father Present=0					
Affected*Earthquake	-0.0302	-0.134	0.0573	-0.195**	-0.130
	(0.114)	(0.106)	(0.106)	(0.0945)	(0.0926)
Observations	7,985	8,624	7,111	8,548	6,985
$Adj - R^2$	0.239	0.152	0.142	0.202	0.101
P value A=B	0.228	0.887	0.253	0.593	0.817

Notes: Each column represents a separate regression. The dependent variable is the test score measured in standard deviations. Peabody is the Peabody Test Score, Battelle is the Battelle Test Score and CBCL is the Child Behavior Check List measure. Observations are at the individual/municipality level. Short- and Medium-term mean measured two and seven years later. Standard errors clustered at the municipal level appear in parentheses—number of clusters: 145. The estimates use sampling weights representative at the national level. Child/Mother/HH characteristics include gender, age at the moment of the test (in months), birth order, mothers' age and schooling, father present, number member of the household, prior mental health condition for mother, father and relative. All specifications contain cohort of birth and municipality fixed effects. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table C.37: The Impact of the Earthquake on Cognitive and Non-Cognitive Development by Fathers' Schooling

	(1)	(2)	(3)	(4)	(5)
		Short-Term		Medium-Term	
	Battelle	Peabody	CBCL1	Peabody	CBCL2
Panel A: Fathers' Educ≤12					
Affected*Earthquake	-0.0529	-0.158	0.0867	-0.212**	-0.0953
	(0.125)	(0.115)	(0.125)	(0.0928)	(0.0998)
Observations	5,898	6,495	5,219	6,431	5,487
$Adj - R^2$	0.238	0.133	0.130	0.186	0.0828
Panel B: Fathers' Educ>12					
Affected*Earthquake	-0.0198	-0.126	-0.00241	-0.142	-0.225
	(0.141)	(0.133)	(0.145)	(0.135)	(0.250)
Observations	2,522	2,669	2,261	2,656	1,998
$Adj - R^2$	0.253	0.154	0.122	0.212	0.0997
P value A=B	0.905	0.491	0.058	0.346	0.697

Notes: Each column represents a separate regression. The dependent variable is the test score measured in standard deviations. Peabody is the Peabody Test Score, Battelle is the Battelle Test Score and CBCL is the Child Behavior Check List measure. The sample is reduced since information of education of non-present fathers is not available. Observations are at the individual/municipality level. Short- and Medium-term mean measured two and seven years later. Standard errors clustered at the municipal level appear in parentheses—number of clusters: 145. The estimates use sampling weights representative at the national level. Child/Mother/HH characteristics include gender, age at the moment of the test (in months), birth order, mothers' age and schooling, father present, number member of the household, prior mental health condition for mother, father and relative. All specifications contain cohort of birth and municipality fixed effects. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table C.38: The Impact of the Earthquake on Cognitive and Non-Cognitive Development by 2017 Income

	(1)	(2)	(3)	(4)	(5)
		Short-Term			Medium-Term
	Battelle	Peabody	CBCL1	Peabody	CBCL2
Panel A: Income≤41,979					
Affected*Earthquake	-0.0999	-0.158*	0.127	-0.183***	-0.191
	(0.0976)	(0.0833)	(0.129)	(0.0621)	(0.116)
Observations	6,716	7,356	5,988	7,273	6,070
$Adj - R^2$	0.239	0.134	0.137	0.174	0.0949
Panel B: Income>41,979					
Affected*Earthquake	-0.0625	-0.127	0.124	-0.168*	-0.0610
	(0.118)	(0.113)	(0.105)	(0.0936)	(0.107)
Observations	6,322	6,815	5,644	6,762	5,475
$Adj - R^2$	0.221	0.143	0.138	0.196	0.126
P value A=B	0.737	0.698	0.988	0.76	0.429

Notes: Each column represents a separate regression. The dependent variable is the test score measured in standard deviations. Peabody is the Peabody Test Score, Battelle is the Battelle Test Score and CBCL is the Child Behavior Check List measure. 41,979 Chilean pesos is the median household income percapita. Observations are at the individual/municipality level. Short- and Medium-term mean measured two and seven years later. Standard errors clustered at the municipal level appear in parentheses—number of clusters: 145. The estimates use sampling weights representative at the national level. Child/Mother/HH characteristics include gender, age at the moment of the test (in months), birth order, mothers' age and schooling, father present, number member of the household, prior mental health condition for mother, father and relative. All specifications contain cohort of birth and municipality fixed effects. * p < 0.1, ** p < 0.05, *** p < 0.01.

Table C.39: The Impact of the Earthquake on Birth Outcomes from Vital Records

	(1)	(2)	(3)	(4)	(5)
	Weight at Birth	LBW	Height at Birth	Birth weeks	Preterm birth
In Utero*Earthquake	-0.0036	-0.0002	-0.125***	-0.0052	-0.0006
	(0.0038)	(0.0014)	(0.0281)	(0.0134)	(0.0014)
Average	3.3201	0.0614	49.33	38.47	0.075
Observations	2,428,467	2,428,467	2,428,188	2,428,509	2,428,509
$Adj - R^2$	0.00853	0.00138	0.0179	0.0150	0.00295

Notes: Each column represents a separate regression. Observations are at the individual/municipality level. Treated group is children in utero at the moment of the earthquake and the control group is the children born before. LBW: Low Birth Weight (<2.5kg). Preterm birth: less than 37 weeks of gestation. Standard errors clustered at the municipal level appear in parentheses—number of clusters: 145 in ELPI and 344 in Vital Records. Vital Records Child/Mother/HH characteristics include mother's age and schooling, father present, number of children of the household, urban, and mother single status. All specifications contain cohort of birth and municipality fixed effects. * p < 0.1, ** p < 0.05, *** p < 0.01.

Table C.40: The Impact of the Earthquake on Mothers' Risky Behaviors using as a control group only the children 7 months old and older

	Risky Behaviors				
	(1)	(2)	(3)	(4)	(5)
	All	Smoke		Alcohol	Drug
		Extensive	Unconditional Intensity		
Treated*Earthquake	0.0469*	0.042*	-0.2186	0.0252	0.00901
	(0.0271)	(0.0251)	(0.4487)	(0.4487)	(0.00869)
Average	0.2180	0.1387	5.8650	0.1092	0.0140
Observations	10,361	10,361	10,361	10,361	10,361
$Adj - R^2$	0.0407	0.0507	0.1971	0.0240	0.0209
Treated	In Utero 0 to 6 mo	In Utero 0 to 6 mo	In Utero 0 to 6 mo	In Utero 0 to 6 mo	In Utero 0 to 6 mo
Control	$>$ 6 mo	$>$ 6 mo	$>$ 6 mo	$>$ 6 mo	$>$ 6 mo

Notes: Each column represents a separate regression. Column 3 is a Poisson model estimated using PPML regression and its coefficient is interpreted as a percentage change. Smoke "Unconditional Intensity" margin measures the number of cigarettes per month. All other columns measure the probability of smoke, drink or consume recreational drugs. All means that at least one of these: smoke, drink or consume drugs. The treated group corresponds to children who were in utero and up to six months old at the moment of the earthquake and the control group to children who were at least seven months old or who were born after the earthquake. The dependent variable is a dummy variable that takes the value 1 if the mother behaves in that specific way. Observations are at the individual/municipality level. Standard errors clustered at the municipal level appear in parentheses, number of clusters: 145. The estimates use sampling weights representative at the national level. Mother/HH characteristics include mothers' age and schooling, father present, number member of the household, prior mental health condition for mother, father and relative. All specifications contain cohort of birth and municipality fixed effects. Data comes from the ELPI survey. * p $<$ 0.1, ** p $<$ 0.05, *** p $<$ 0.01.

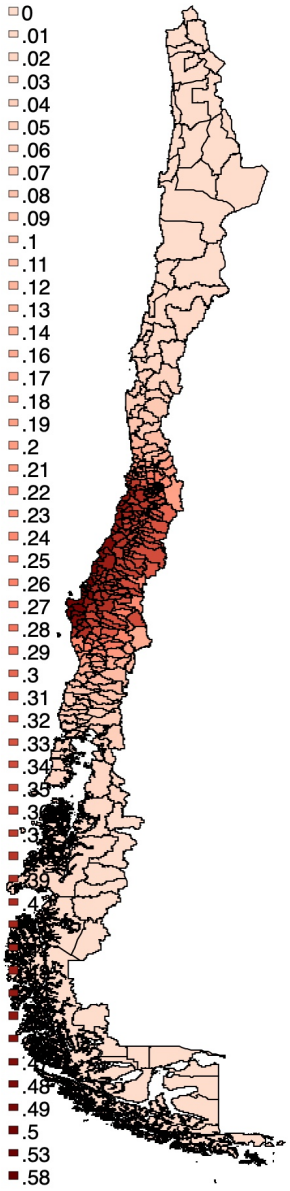
Table C.41: The Impact of the Earthquake on Smoking: Positive Smoking and Unconditional Intensity margin excluding outliers

	Smoke>0		0≤Smoke<70	
	(1)	(2)	(3)	(4)
Treated*Earthquake	-0.3600	-0.3295	0.8253**	0.5852*
	(0.3781)	(0.4089)	(0.4205)	(0.3528)
Average	41.79	42.91	1.91	8.95
Observations	1,815	1,418	15,213	10,165
$Adj - R^2$	0.26	0.2354	0.1574	0.1184
Control	> 6 mo Unborn	> 6 mo	> 6 mo Unborn	> 6 mo

Notes: Each column represents a separate regression. All columns use a Poisson model estimated using PPML regression and its coefficient is interpreted as a percentage change. $Smoke > 0$ is the sample of mothers reporting positive consumption at the time of the survey. 0≤Smoke<70 is the sample that blends extensive and intensive margin excluding the 99th percentile of smokers. The treated group corresponds to children who were in utero and up to six months old at the moment of the earthquake and the control group to children who were at least seven months old and/or who were born after the earthquake. The dependent variable is the number of cigarettes per month. 70 cigarettes per month is the 99 percentile. Observations are at the individual/municipality level. Standard errors clustered at the municipal level appear in parentheses—number of clusters: 145. The estimates use sampling weights representative at the national level. Mother/HH characteristics include mothers' age and schooling, father present, number member of the household, prior mental health condition for mother, father and relative. All specifications contain cohort of birth and municipality fixed effects. Data comes from the ELPI survey. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

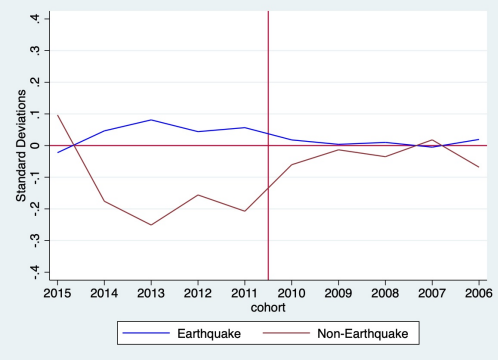
D Appendix: Figures

Figure D.1: Intensities of the 2010 Chilean Earthquake at municipality level for the country

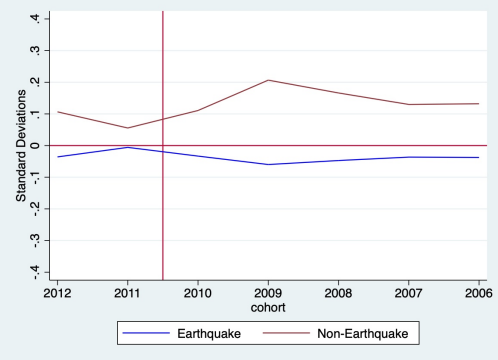


Note: The figures displays the Peak Ground Acceleration intensities in all municipalities.

Figure D.2: Cohort Raw Trends: Medium-term Development



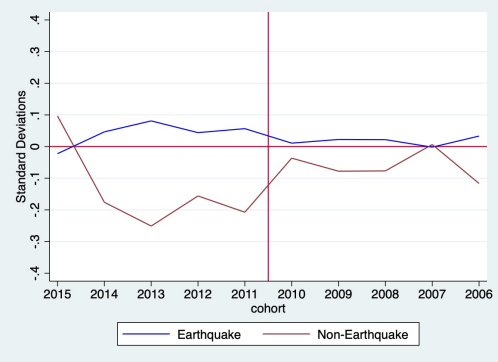
(a) Peabody



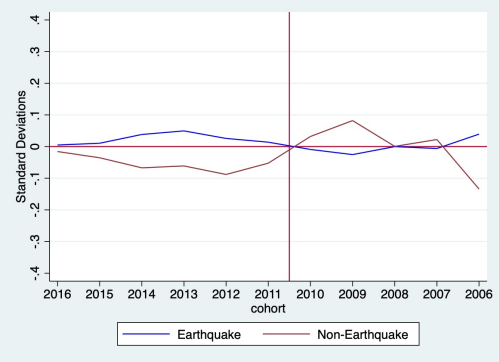
(b) Battelle

Note: This figure displays the trends on short-term cognitive and non-cognitives outcomes among cohorts born from 2006 to 2016. Data originates from the representative Encuesta Longitudinal de la Primera Infancia (ELPI), using sampling weights to ensure representativeness. Municipalities are classified as Earthquake Impacted (treatment) or Non-Earthquake impacted (control). The vertical line at 2010 indicates the occurrence of the 8.8 magnitude earthquake in Chile, allowing for comparison of behavioral shifts following disaster exposure.

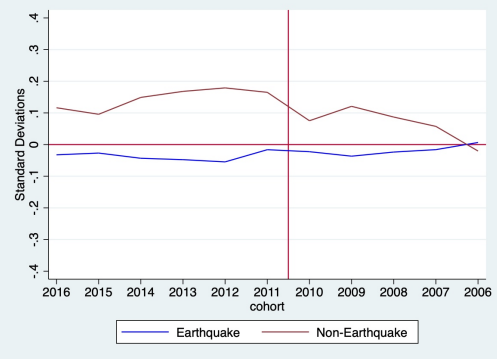
Figure D.3: Cohort Raw Trends: Short-term Development



(a) Peabody



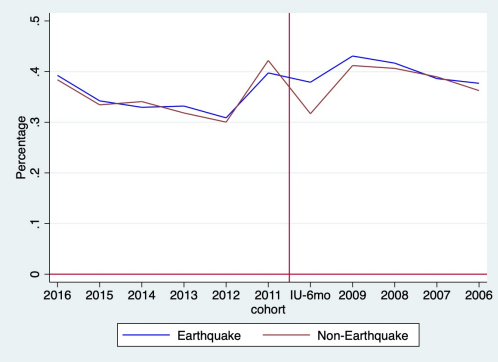
(b) Battelle



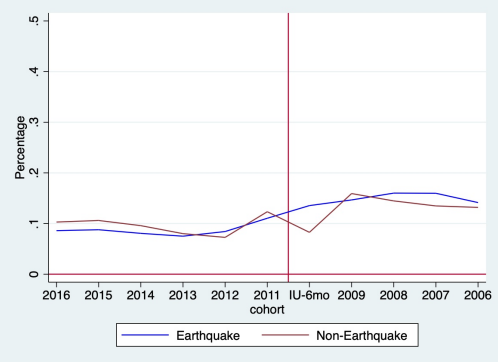
(c) CBCL

Note: This figure displays the trends on medium-term cognitive and non-cognitives outcomes among cohorts born from 2006 to 2016. Data originates from the representative Encuesta Longitudinal de la Primera Infancia (ELPI), using sampling weights to ensure representativeness. Municipalities are classified as Earthquake Impacted (treatment) or Non-Earthquake impacted (control). The vertical line at 2010 indicates the occurrence of the 8.8 magnitude earthquake in Chile, allowing for comparison of behavioral shifts following disaster exposure.

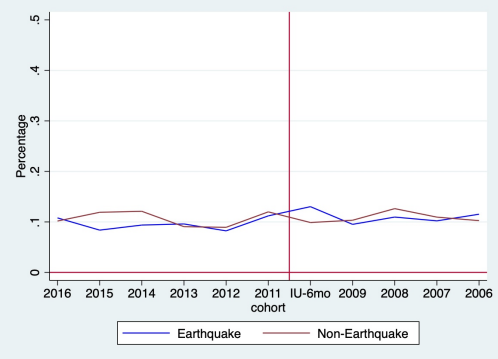
Figure D.4: Risky Behaviors Raw Trends



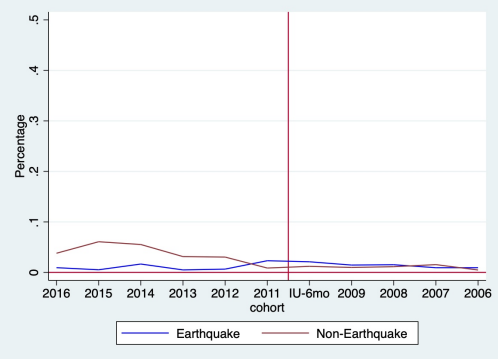
(a) Risky Behaviors



(b) Smoking



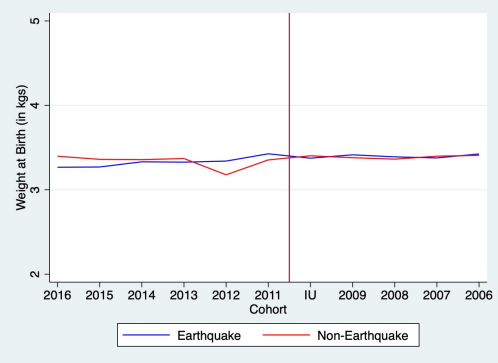
(c) Drinking



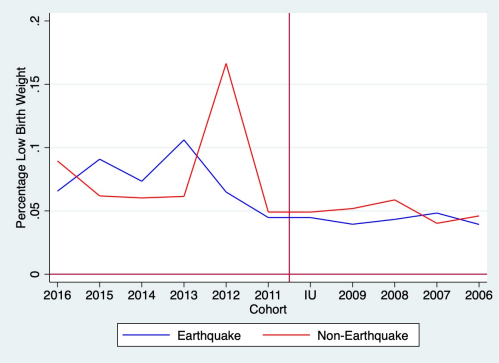
(d) Recreational Drugs

Note: This figure displays the trends in the probability of maternal risky behaviors (at least one of smoking, drinking, and recreational drug use), smoking, driking and consumption of recreational drugs among cohorts born from 2006 to 2016. Data originates from the representative Encuesta Longitudinal de la Primera Infancia (ELPI), using sampling weights to ensure representativeness. Municipalities are classified as Earthquake Impacted (treatment) or Non-Earthquake impacted (control). *IU-6mo* is the cohort of children who were in utero and up to 6 months old, the treated grupo for the risky behavior analysis. The vertical line at 2010 indicates the occurrence of the 8.8 magnitude earthquake in Chile, allowing for comparison of behavioral shifts following disaster exposure.

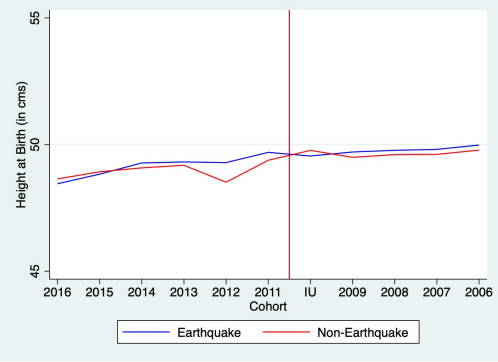
Figure D.5: Birth Outcomes Raw Trends



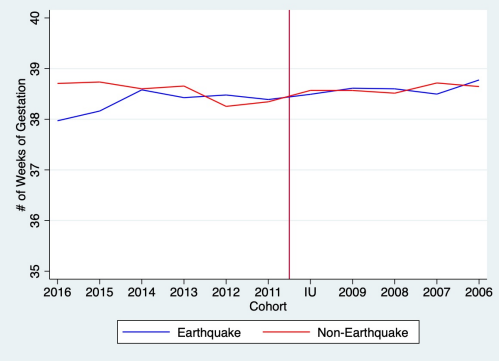
(a) Weight at Birth



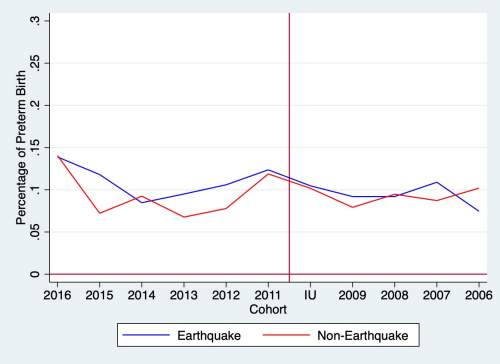
(b) Low Birth Weight



(c) Height at Birth



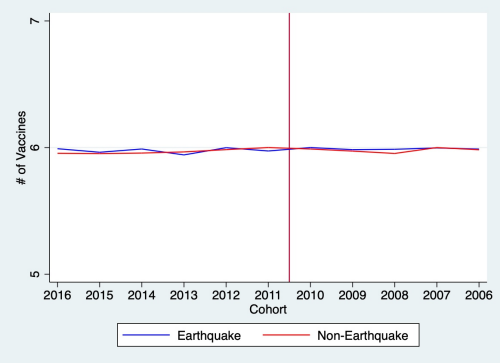
(d) Gestational Weeks



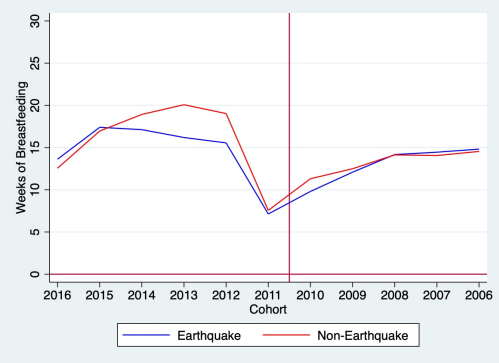
(e) Pr(Preterm Birth)

Note: This figure displays the trends on birth outcomes among cohorts born from 2006 to 2016. Data originates from the representative Encuesta Longitudinal de la Primera Infancia (ELPI), using sampling weights to ensure representativeness. Municipalities are classified as Earthquake Impacted (treatment) or Non-Earthquake impacted (control). *IU* is the in utero cohort. The vertical line at 2010 indicates the occurrence of the 8.8 magnitude earthquake in Chile, allowing for comparison of behavioral shifts following disaster exposure..

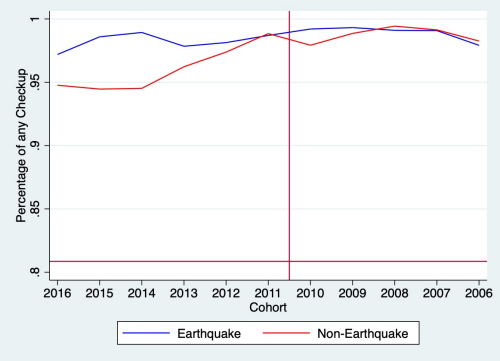
Figure D.6: Health Outcomes Raw Trends



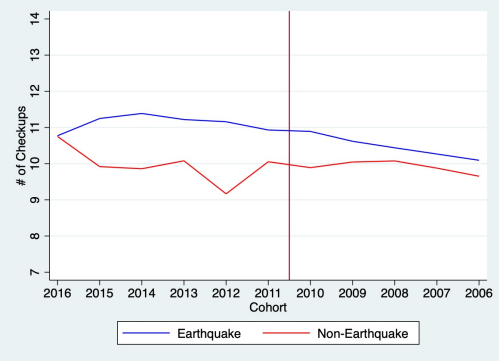
(a) Number of Vaccines



(b) Months of Breastfeeding



(c) Pr(Checkup)



(d) Number of Checkups

Note: This figure displays the trends on health outcomes among cohorts born from 2006 to 2016. Data originates from the representative Encuesta Longitudinal de la Primera Infancia (ELPI), using sampling weights to ensure representativeness. Municipalities are classified as Earthquake Impacted (treatment) or Non-Earthquake impacted (control). The vertical line at 2010 indicates the occurrence of the 8.8 magnitude earthquake in Chile, allowing for comparison of behavioral shifts following disaster exposure.